WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

ZHIJIE FAN, JI LI, AND DMITRIY ZANIN

Abstract.

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1. Introduction

1.1. **Background and motivation.** [Zhijie will write the background in the Dunkl setting.]

[Comment: The following Besov space never appear in any literature, and it is unclear the equivalence (most likely not equivalent) with any Besov space studied in any literature.]

[Question 1: what is a suitable name for our Besov space?]

[Question 2: what is the relationship between these Besov spaces?]

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Definition 1.1. Suppose $1 < p < \infty$. We say that a function $f \in L^p_{\mathrm{loc}}(\mathbb{R}^N, dw)$ belongs to Dunkl Besov space $B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)$ if

$$\|f\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N}\int_{\mathbb{R}^N}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|f(x)-f(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\right)^{1/p} < +\infty.$$

Theorem 1.2. *Suppose $N \geq 1$, $\max\{2, N\} < p < \infty$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. For any $j \in \{1, 2, ..., N\}$, we have $[b, R_{\mathrm{Dunkl},j}] \in S^p$ if and only if $b \in B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)$. Moreover, we have*

$$\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p} \approx \|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)}.$$

Theorem 1.3. *Suppose $N \geq 2$, $0 < p \leq N$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. For any $j \in \{1, 2, ..., N\}$, we have $[b, R_{\mathrm{Dunkl},j}] \in S^p$ if and only if b is a constant on $\mathbb{R}^N$.*

[Question 1: how to obtain

$$\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p} \lesssim \|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)}.$$

when $N = 1$ and $1 < p < 2$?]

[Comment: Suppose $N \geq 2$, then combining Theorems 1.2 and 1.3, we obtain a full range of Schatten-p characterization. However, when $N = 1$, the upper bound in Proposition 4.5 only works for $p > 2$, though we obtain the lower bound in Proposition 5.9 for $p > 1$.]

[Question 2: How to establish

$$\|[b, R_{\mathrm{Dunkl},j}]\|_{S^{N,\infty}} \approx \|b\|_{\mathrm{something}}?$$

Comment: I even can't guess what the form of "something" above should be. The appropriate forms of Sobolev spaces and oscillation spaces are both unclear.]

1.2. **Difficulty and strategy.** [Zhijie will add this later]

1.3. **Notation.** Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N, p, and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda\,\ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p, which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ-measurable set E by

$$(f)_E := \fint_E f(x)d\mu(x) := \frac{1}{\mu(E)}\int_E f(x)d\mu(x). \tag{1.1}$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. **Harmonic analysis in the Dunkl setting.** Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle\cdot,\cdot\rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v\in\mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2\frac{\langle x,v\rangle}{\|v\|^2}v.$$

A finite set $R\subset\mathbb{R}^N\backslash\{0\}$ is called a root system if $\sigma_v(R)=R$ for every $v\in R$. A root system R is called a normalized reduced root system if $\|v\|^2=2$ for every $v\in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v\in R}$ is called the Weyl group (reflection group) of the root system R. Let $k:R\to[0,\infty)$ be a multiplicity function, which is a G-invariant function fixed throughout the paper. We will denote by $\mathcal{O}(x)=\{\sigma(x):\sigma\in G\}$ the G-orbit of a point $x\in\mathbb{R}^N$, and $\mathcal{O}(E)=\bigcup_{x\in E}\mathcal{O}(x)$ the G-orbit of $E\subset\mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v\in R}|\langle x,v\rangle|^{k(v)}dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x,r):=\{y\in\mathbb{R}^N:\|x-y\|<r\}$ be the Euclidean ball centred at x with radius $r>0$. Let $\mathbf{N}:=N+\sum_{v\in R}k(v)$. Note that

$$w(B(tx,tr))=t^{\mathbf{N}}w(B(x,r)),\ \text{for } x\in\mathbb{R}^N,\ t,r>0.$$

Furthermore,

$$w(B(x,r))\sim r^N\prod_{v\in R}(|\langle x,v\rangle|+r)^{k(v)}, \tag{2.1}$$

where the implicit constant only depends on N,R and k. It follows from (2.1) that there exists a constant $C\geq 1$ such that for every $x\in\mathbb{R}^N$,

$$C^{-1}\left(\frac{r_2}{r_1}\right)^N\leq\frac{w(B(x,r_2))}{w(B(x,r_1))}\leq C\left(\frac{r_2}{r_1}\right)^{\mathbf{N}},\ \text{for } 0<r_1<r_2, \tag{2.2}$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x)+\sum_{v\in R}\frac{k(v)}{2}\langle v,\xi\rangle\frac{f(x)-f(\sigma_v(x))}{\langle v,x\rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ. Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\mathrm{Dunkl}}f(x) := \sum_{j=1}^N T_{e_j}^2 f(x)=\Delta_{\mathbb{R}^N}f(x)+\sum_{v\in R}k(v)\delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v,x\rangle}-\frac{f(x)-f(\sigma_v(x))}{\langle v,x\rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\mathrm{Dunkl},j} := T_{e_j}(-\Delta_{\mathrm{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$\left\{x \in \mathbb{R}^N : \langle x, v\rangle \neq 0 \quad \textit{for all } v \in R\right\}. \tag{2.3}$$

Define $d(x, y) := \min_{\sigma\in G} \|x - \sigma(y)\|$ as the distance between two G-orbits $\mathcal{O}(x)$ and $\mathcal{O}(y)$. Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma \tag{2.4}$$

for some closed Weyl chamber Γ.

It is clear that $\mathcal{O}(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(\mathcal{O}(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = \mathcal{O}(B(x, r))$.

Denote by $K_{\mathrm{Dunkl},j}(x, y)$ the integral kernel of $R_{\mathrm{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \dots, N\}$ and every x, y with $d(x, y) \neq 0$,*

$$|K_{\mathrm{Dunkl},j}(x, y)| \leq C\frac{d(x, y)}{\|x - y\|}\frac{1}{w(B(x, d(x, y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

Lemma 2.3. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\mathrm{Dunkl},j}(x, y) \geq c_{R,k}\frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\mathrm{Dunkl},j}(x, y) \leq c_{R,k}\frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The lower bound is identical to that in [5, Theorem 1.2]. □

2.2. **Schatten class.** [Zhijie will add this later]

2.3. **Systems of dyadic cubes.** At the beginning of this subsection, we recall the standard notion of dyadic cube systems in $\mathbb{R}^N$. [Zhijie will write more background]

Definition 2.4. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathscr{D} := \bigcup_{k\in\mathbb{Z}} \mathscr{D}_k$, $\mathscr{D}_k := \{Q^k_\alpha : \alpha \in \mathscr{A}_k\}$, of Borel sets $Q^k_\alpha \subseteq X$ is called *a system of dyadic cubes over X* if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha\in\mathscr{A}_k} Q^k_\alpha;$$

(ii) If $\ell \geq k$, then either $Q^\ell_\beta \subseteq Q^k_\alpha$ or $Q^k_\alpha \cap Q^\ell_\beta = \varnothing$;

(iii) For each (k,α) and each $\ell \le k$, there exists a unique β such that $Q^k_\alpha \subseteq Q^\ell_\beta$;

(iv) For each (k,α) there exist 2^N indices β such that

$$Q^{k+1}_\beta \subseteq Q^k_\alpha, \text{ and } Q^k_\alpha = \bigcup_{\substack{Q\in\mathscr{D}_{k+1}\\ Q\subseteq Q^k_\alpha}} Q;$$

(v) For each (k,α), one has

$$B_X(x^k_\alpha, A_1 2^{-k}) \subseteq Q^k_\alpha \subseteq B_X(x^k_\alpha, A_2 2^{-k}) =: B(Q^k_\alpha)$$

for some positive constants A_1, A_2 depending only on the geometry of X, where we set $B_X(x,r) := B(x,r)\cap X$ for all $x\in X$ and $r>0$.

(vi) If $\ell \ge k$ and $Q^\ell_\beta \subseteq Q^k_\alpha$, then

$$B(Q^\ell_\beta) \subseteq B(Q^k_\alpha).$$

Each Q^k_α is called a *dyadic cube of generation* k, with center $x^k_\alpha \in Q^k_\alpha$ and side-length $\ell(Q^k_\alpha) = 2^{-k}$.

Notation 2.5. *For every dyadic cube $Q\in\mathscr{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.*

Notation 2.6. *Let*

$$\mathbb{D}^0 := \bigcup_{k\in\mathbb{Z}} \mathbb{D}^0_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1+1)2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N+1)2^{-k}),$$

with $(m_1,\dots,m_N)\in\mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_{\Gamma,k}$$

be the system of dyadic cubes over a fixed chamber Γ, whose construction will be given in Corollary 3.7.

[Zhijie will add the following later]

1. Explain that why don't we use the version of Conde-Alonso;

2. Add background about collection of systems of dyadic cubes over spaces of homogeneous type, and explain that we can certain use this concept to do the required thing, but why don't we use this concept

---

Lemma 2.7. *There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \le r_B < \frac{2^{-k-1}}{3}$ for some $k\in\mathbb{Z}$, there exists a cube $Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu_k$ such that*

$$B \subset Q \subset 12\sqrt{N}B.$$

Proof. • **Step 1.** *Recall the one-dimensional setting given in* [13, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.6). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k\in\mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \Big\{ \Big[\frac{m}{2^k} + \delta + \sum_{j=(k/2)+1}^{0} 2^{-2j},\ \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^{0} 2^{-2j} \Big) \ \Big|\ m \in \mathbb{Z} \Big\}.$$

These choices together with the nestedness property property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

(1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
(2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^{0} 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
(3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^{0} 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a-b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J. Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• **Step 2.** *We extend the construction to* $\mathbb{R}^N$ *(*$N \geq 1$*).*

For $\vartheta = (\vartheta_1, \vartheta_2, \ldots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^{N} I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i-th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12 r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \ldots, \vartheta_N)$, and let $Q = \prod_{j=1}^{N} I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12 r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N}\ell(Q) \leq 12\sqrt{N} r_B.$$

This implies that $Q \subset B(x_B, 12\sqrt{N} r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \ldots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.7. □

2.4. **Haar Basis.** [Zhijie will add more background later]

Given a system of dyadic cubes $\mathscr{D}$ over $\mathbb{R}^N$, we now recall the standard way to construct an orthonormal Haar system corresponding to $\mathscr{D}$. For each $k \in \mathbb{Z}$ and $Q \in \mathscr{D}_k$, denote by $Ch(Q) := \{R \in \mathscr{D}_{k+1} : R \subseteq Q\}$. Moreover, we let

$$h_Q^0 := w(Q)^{-1/2} \chi_Q.$$

Fix an enumeration $\{Q_i\}_{i=1}^{2^N}$ of $Ch(Q)$. Let

$$\hat{Q}_i := \bigcup_{j=i}^{2^N} Q_j$$

and

$$h_Q^i := \sqrt{\frac{w(Q_i)w(\hat{Q}_{i+1})}{w(\hat{Q}_i)}}\left(\frac{\chi_{Q_i}}{w(Q_i)} - \frac{\chi_{\hat{Q}_{i+1}}}{w(\hat{Q}_{i+1})}\right), \quad i = 1, 2, \ldots, 2^N - 1.$$

Lemma 2.8. *For any $1 < p < \infty$ and $f \in L^p(\mathbb{R}^N, dw)$, we have*

$$f(x) = \sum_{Q\in\mathscr{D}} \sum_{i=1}^{2^N-1} \langle f, h_Q^i\rangle_{L^2(\mathbb{R}^N,dw)} h_Q^i(x),$$

where the sum converges both in the $L^p(\mathbb{R}^N, dw)$ norm and pointwise almost everywhere.

Lemma 2.9. *The family of Haar functions $\{h_Q^i\}_{Q\in\mathscr{D},0\le i\le 2^N-1}$ has the following properties:*

(i) *h_Q^i is supported on Q for $i \in \{0, 1, \ldots, 2^N - 1\}$;*
(ii) *h_Q^i is constant on each $R \in Ch(Q)$ for $i \in \{0, 1, \ldots, 2^N - 1\}$;*
(iii) *$\int_{\mathbb{R}^N} h_Q^i(x)dw(x) = 0$, for $i \in \{1, 2, \ldots, 2^N - 1\}$;*
(iv) *$\langle h_Q^i, h_Q^j\rangle_{L^2(\mathbb{R}^N,dw)} = 0$ for $i \neq j$, i, $j \in \{1, 2, \ldots, 2^N - 1\}$;*
(v) *The collection $\{h_Q^i\}_{i=0}^{2^N-1}$ is an orthogonal basis for the vector space $V(Q)$ of all functions on Q that are constant on each sub-cube $R \in Ch(Q)$.*
(vi) *$\|h_Q^i\|_{L^p(\mathbb{R}^N,dw)} \approx w(Q)^{\frac{1}{p}-\frac{1}{2}}$ for $1 \le p \le \infty$ and $i \in \{1, 2, \ldots, 2^N - 1\}$;*

2.5. **Nearly weakly orthonormal sequences.** Rochberg and Semmes [15] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [4, 8, 14, 2, 3, 12, 15, 18, 17]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathscr{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q\in\mathscr{D}}$ is an NWO sequence if

(1) $\operatorname{supp} e_Q \subseteq cQ$ *for some positive constant c independent of Q*
(2) $\|e_Q\|_{L^s(\mathbb{R}^N,dw)} \le w(Q)^{1/s-1/2}$ *for some $s > 2$.*

Lemma 2.10. *Let $1 < p < \infty$, and let $\mathscr{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. For any compact operator T on $L^2(\mathbb{R}^N, dw)$, we have*

$$\Big(\sum_{Q\in\mathscr{D}} |\langle Te_Q, f_Q\rangle_{L^2(\mathbb{R}^N,dw)}|^p\Big)^{1/p} \le C_{N,p,w}\|T\|_{S^p},$$

where $\{e_Q\}_{Q\in\mathscr{D}}$ and $\{f_Q\}_{Q\in\mathscr{D}}$ are NWO sequences.

Proof. This property can be found in [15, (1.10)] (see also [8, Section 7]). □

Lemma 2.11. *Let $0 < p < \infty$, and let $\mathscr{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that*

$$A = \sum_{Q \in \mathscr{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathscr{D}}$ and $\{f_Q\}_{Q \in \mathscr{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathscr{D}}$ is a sequence of scalars. We have

$$\|A\|_{S^p} \leq C_{N,p,w} \|\{\lambda_Q\}_Q\|_{\ell^p}. \tag{2.5}$$

Proof. This property can be found in [15, (1.9)] (see also [8, Section 7]). □

3. Analytic tools for Weyl chamber geometry

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \operatorname{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [6, Definition 2.7] *A connected component of*

$$\left\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \quad \textit{for all } v \in R\right\} \tag{3.1}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive *and* negative *roots by*

$$R_+ := \{\alpha \in R : \ \langle x, \alpha \rangle > 0 \textit{ for all } x \in \Gamma_+\}, \qquad R_- := -R_+.$$

Definition 3.2. [6, Definition 2.9] *A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as*

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \textit{with } \lambda_1, \lambda_2 \geq 1 \textit{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. *With the notation given in Definition 3.2, Δ is a basis of W. If we denote by $\Delta = \{\alpha_1, \alpha_2, \ldots, \alpha_{N_0}\}$, then the open and closed chambers can be written as*

$$\Gamma_+ = \left\{x \in \mathbb{R}^N : \ \langle x, \alpha_i \rangle > 0, \ i = 1, \ldots, N_0\right\}, \tag{3.2}$$

$$\Gamma = \left\{x \in \mathbb{R}^N : \ \langle x, \alpha_i \rangle \geq 0, \ i = 1, \ldots, N_0\right\}, \tag{3.3}$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+. By [6, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha\rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \operatorname{span}(\Delta)$ and hence,

$$W = \operatorname{span}(R) \subset \operatorname{span}(\Delta).$$

The reverse inclusion $\operatorname{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\operatorname{span}(\Delta) = W$. On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W, and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [6, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+, note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ, specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. □

Notation 3.4. *With the notation given in Lemma 3.3, denote by*

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ.

Lemma 3.5. *Each Weyl chamber* Γ *admits the decomposition*

$$\Gamma = K(\Gamma) \times W^\perp, \qquad K(\Gamma) := \{w \in W : \ \langle w, \alpha\rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i\rangle = \langle w, \alpha_i\rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. □

Corollary 3.6. *Let* $\Phi_{W^\perp}$ *be any linear isomorphism from* $W^\perp$ *to* $\mathbb{R}^{N-N_0}$*, and let*

$$\Phi_W : W \longrightarrow \mathbb{R}^{N_0}, \qquad \Phi_W(w) := \big(\langle w, \alpha_1\rangle, \dots, \langle w, \alpha_{N_0}\rangle\big),$$

$$\Phi : \mathbb{R}^N = W \oplus W^\perp \longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, \qquad \Phi(w+z) := \big(\Phi_W(w),\, \Phi_{W^\perp}(z)\big).$$

Then Φ *is a linear isomorphism and*

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \ \langle w, \alpha_i\rangle \geq 0, \ i = 1, \dots, N_0\}.$$

Since Δ is a basis of W, the Gram matrix $G = (\langle \alpha_i, \alpha_j\rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \longrightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W\big(K(\Gamma)\big) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi\big(K(\Gamma) \times W^\perp\big) = \Phi_W\big(K(\Gamma)\big) \times \Phi_{W^\perp}\big(W^\perp\big) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. □

Corollary 3.7. *Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ.*

Proof. Let $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}^0_{\Omega,k}$ to be the family of half-open cubes

$$Q = \prod_{i=1}^{N} I_i, \qquad I_i = \big[m_i 2^{-k}, (m_i+1)2^{-k}\big),$$

where $m_i \in \mathbb{Z}_+$ for $1 \le i \le N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \le N$. Set

$$\mathbb{D}^0_\Omega := \bigcup_{k\in\mathbb{Z}} \mathbb{D}^0_{\Omega,k}.$$

It is standard that $\mathbb{D}^0_\Omega$ is a system of dyadic cubes over Ω.

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \big\{\Phi^{-1}(Q) : Q \in \mathbb{D}^0_{\Omega,k}\big\}, \qquad \mathbb{D}_\Gamma := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_\Gamma$ is a system of dyadic cubes over Γ. □

Notation 3.8. *Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define*

$$\mu(E) := w\big(\Phi^{-1}(E)\big). \tag{3.4}$$

Lemma 3.9. *We have*

$$d\mu(u,z) = \phi(u)\,du\,dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v\in R_+} \Big(\sum_{i=1}^{N_0} a_{v,i}\,u_i\Big)^{2k(v)}. \tag{3.5}$$

where for each $v \in R_+$, the coefficients $a_{v,i} \ge 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i}\alpha_i$ and satisfy $(a_{v,1},\dots,a_{v,N_0}) \ne \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω,

$$\int_\Omega f(u,z)\,d\mu(u,z) = \int_\Gamma f\big(\Phi(x)\big)\,dw(x).$$

We make the linear change of variables $(u,z) = \Phi(x)$, so that $x = \Phi^{-1}(u,z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1}\,du\,dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i\rangle$ for $i = 1,\dots,N_0$. This implies that

$$\langle x, v\rangle = \Big\langle x, \sum_{i=1}^{N_0} a_{v,i}\alpha_i\Big\rangle = \sum_{i=1}^{N_0} a_{v,i}\langle x,\alpha_i\rangle = \sum_{i=1}^{N_0} a_{v,i}u_i. \tag{3.6}$$

Substituting these identities into the previous integral, we obtain

$$\int_\Omega f(u,z)\,d\mu(u,z) = \int_\Gamma f\big(\Phi(x)\big)\Big(\prod_{v\in R_+}|\langle x,v\rangle|^{2k(v)}\Big)\,dx$$
$$= \int_\Omega f(u,z)\,|\det(\Phi)|^{-1}\prod_{v\in R_+}\Big(\sum_{i=1}^{N_0} a_{v,i}u_i\Big)^{2k(v)}\,du\,dz.$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. □

Lemma 3.10. *Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1,\dots,\alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and*

$$\Gamma^{(k)} := \sigma_{\alpha_k}\big(\Gamma^{(k-1)}\big), \qquad k = 1,\dots,M,$$

then the following hold:

(1) *$\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1,\dots,M$;*
(2) *$\Gamma^{(M)} = \Gamma_1$.*

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1,\beta_2,\dots,\beta_{N_0}\}$ be the simple system associated with Γ_2. By [6, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1},\sigma_{\beta_2},\dots,\sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1,\dots,i_M \in \{1,\dots,N_0\}$ such that

$$g = \sigma_{\beta_{i_1}}\sigma_{\beta_{i_2}}\cdots\sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \mathrm{Id}, \qquad w_k := \sigma_{\beta_{i_1}}\cdots\sigma_{\beta_{i_k}} \quad (k = 1,\dots,M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \qquad (k = 0,1,\dots,M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1,\dots,M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

• **Step 1.** *Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.*

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w\,\sigma_\beta\,w^{-1} = \sigma_{w(\beta)}, \qquad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1}\sigma_{\beta_{i_k}}w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1}\sigma_{\beta_{i_k}}w_{k-1}^{-1}\big(w_{k-1}(\Gamma_2)\big) = w_{k-1}\sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

• **Step 2.** *Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.*

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \ \langle x,\gamma\rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x,\, \beta\rangle = \langle x,\, w_{k-1}^{-1}\beta\rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \Big\{x \in \mathbb{R}^N :\ \langle x,\, w_{k-1}(\beta_j)\rangle \geq 0 \text{ for } j = 1, \dots, N_0\Big\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta\big(w_{k-1}(\Gamma_2)\big) = w_{k-1}\big(\Delta(\Gamma_2)\big) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. □

4. Proof of upper bound

Let (X, μ) be a measure space and let $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ denote the mixed-norm space consisting of all $\mu \times \mu$-measurable functions $f : X \times X \to \mathbb{C}$ such that

$$\int_X \|f(x, \cdot)\|^p_{L_{q,\infty}(X,\mu)} d\mu(x), \int_X \|f(\cdot, y)\|^p_{L_{q,\infty}(X,\mu)} d\mu(y) < \infty.$$

We equip $L_p(L_{q,\infty})^{\text{symm}}(X \times X, \mu \times \mu)$ with its natural norm.

Lemma 4.1. *Let (X, μ) be a measure space and let $2 < p < \infty$. Suppose that $K \in L_p(L_{p',\infty})^{\text{symm}}(X \times X, \mu \times \mu)$. The integral operator*

$$T_K f(x) := \int_X K(x, y) f(y) d\mu(y)$$

belongs to $S_{p,\infty}(L_2(X, d\mu))$. Moreover, we have

$$\|T_K\|_{S_{p,\infty}(L_2(X,d\mu))} \leq c_p \|K\|_{L_p(L_{p',\infty})^{\text{symm}}(X\times X,\mu\times\mu)}.$$

Proof. The proof is given in [10, Lemma 2] (see also [8, Corollary 11.2]). □

Lemma 4.2. *Let $p > 2$ and let $\frac{1}{q} = 1 - \frac{2}{p}$. We have*

$$\|F_1 F_2\|_{L_p(L_{p',\infty})(X\times X,\mu\times\mu)} \leq 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X\times X,\mu\times\mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

Proof. We clearly have

$$LHS = \Big(\int_X \|F_1(x, \cdot) F_2(x, \cdot)\|^p_{L_{p',\infty}(X,\mu)} d\mu(x)\Big)^{\frac{1}{p}}.$$

Clearly, $\frac{1}{p'} = \frac{1}{p} + \frac{1}{q}$. Thus,

$$\|F_1(x, \cdot) F_2(x, \cdot)\|_{L_{p',\infty}(X,\mu)} \leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_{p,\infty}(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} \leq$$
$$\leq 2^{\frac{1}{p'}} \|F_1(x, \cdot)\|_{L_p(X,\mu)} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)}.$$

Hence,

$$LHS \leq 2^{\frac{1}{pp'}} \Big(\int_X \|F_1(x, \cdot)\|^p_{L_p(X,\mu)} \|F_2(x, \cdot)\|^p_{L_{q,\infty}(X,\mu)} d\mu(x)\Big)^{\frac{1}{p}} \leq$$
$$\leq 2^{\frac{1}{pp'}} \Big(\int_X \|F_1(x, \cdot)\|^p_{L_p(X,\mu)} d\mu(x)\Big)^{\frac{1}{p}} \times \sup_{x\in X} \|F_2(x, \cdot)\|_{L_{q,\infty}(X,\mu)} =$$
$$= 2^{\frac{1}{pp'}} \|F_1\|_{L_p(X\times X,\mu\times\mu)} \|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

□

Lemma 4.3. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $K\in L_p(X\times X)$ and $G\in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FG}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|F\|_{L_p(X\times X,G^2(\mu\times\mu))}\|G\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Proof. By Lemma 4.1,

$$\|T_{FG}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|FG\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Let $\frac{1}{q}=1-\frac{2}{p}$. It follows from Lemma 4.2 that

$$\|F_1F_2\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}\le$$

$$\le 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Setting $F_1=FG^{\frac{2}{p}}$ and $F_2=G^{\frac{1}{q}}$, we obtain

$$\|FG\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}\le$$

$$\le 2^{\frac{1}{pp'}}\|FG^{\frac{2}{p}}\|_{L_p(X\times X,\mu\times\mu)}\|G^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)}=$$

$$=2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,G^2(\mu\times\mu))}\|G\|^{\frac{1}{q}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}=$$

$$=2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,G^2(\mu\times\mu))}\|G\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

□

Lemma 4.4. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $K\in L_p(X\times X)$ and $G\in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FG}\|_{S_p(L_2(X,d\mu))}\le c_{p,G}\|F\|_{L_p(X\times X,G^2(\mu\times\mu))}.$$

Proof. The assertion follows from Lemma 4.3 by real interpolation. □

Proposition 4.5. *Let $p>2$ and $b\in B^p_{\rm Dunkl}(\mathbb{R}^N)$. For every $1\le j\le N$, we have*

$$\|[R_{{\rm Dunkl},j},M_f]\|_{S_p(L_2(\mathbb{R}^N,w(x)dx))}\le c_{R,k}\|f\|_{B^p_{\rm Dunkl}(\mathbb{R}^N)}.$$

Proof. In what follows, let $X=\mathbb{R}^N$ and $d\mu(x)=w(x)dx$. Set

$$F(x,y)=w(B(x,d(x,y)))K_{{\rm Dunkl},j}(x,y)(f(x)-f(y)),\quad x\ne y,$$

$$G(x,y)=\frac{1}{w(B(x,d(x,y)))},\quad x\ne y.$$

By [8, Lemma 11.7], $G\in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. By Lemma 2.2,

$$\|F\|_{L_p(X\times X,G^2(\mu\times\mu))}\le c_R\|f\|_{B^p_{\rm Dunkl}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator $[R_{{\rm Dunkl},j},M_f]$ is FG. The assertion follows now from Lemma 4.4. □

5. Proof of lower bound

it is most inconvenient that integral kernel of $R_{\mathrm{Dunkl},j}$ is not positive. however, the only source of non-positivity is the factor $(x-y)_j$. since triangular truncation operator is bounded on $\mathcal{L}_p$, it follows that, instead of $[R_{\mathrm{Dunkl},j}, M_f]$, we need to consider the operator $T_{j,f}$ whose integral kernel is

$$(f(x)-f(y))\cdot|(x-y)_j|\cdot K(x,y),$$

where K is the POSITIVE integral kernel of $\Delta^{-\frac{1}{2}}$. this should make the life easier.

here is my plan to simplify the argument:

Step 1: verify the assumptions used in Hytonen's paper for every chamber. conclude, using Hytonen theorem, that f is Dunkl-Sobolev in every chamber.

Step 2: fix $g_1, g_2 \in G$ such that $g_1 \neq g_2$. consider integral operator on $L_2(\Gamma\times\Gamma, w(x)dx)$ whose integral kernel is

$$(f(g_1x)-f(g_2y))\cdot|(g_1x-g_2y)_j|\cdot K(g_1x,g_2y), \quad x,y\in\Gamma.$$

if our commutator is in $\mathcal{L}_p$, then so is this integral operator.

on the other hand, if our commutator is in $\mathcal{L}_p$, then the function $f\circ g_2$ is Dunkl-Sobolev in Γ. repeating the argument in Sufficiency part, we conclude that integral operator with integral kernel

$$(f(g_2x)-f(g_2y))\cdot|(g_1x-g_2y)_j|\cdot K(g_1x,g_2y), \quad x,y\in\Gamma,$$

belongs to $\mathcal{L}_p$.

combining the results in the preceding paragraphs, we conclude that integral operator with integral kernel

$$(f(g_1x)-f(g_2x))\cdot|(g_1x-g_2y)_j|\cdot K(g_1x,g_2y), \quad x,y\in\Gamma,$$

belongs to $\mathcal{L}_p$.

Step 3: let $T_{g_1,g_2,j}$ be an integral operator on $L_2(\Gamma\times\Gamma, w(x)dx)$ whose integral kernel is

$$|(g_1x-g_2y)_j|\cdot K(g_1x,g_2y), \quad x,y\in\Gamma.$$

we need to show the following result:

EXPECTED THEOREM: suppose f is Dunkl-Sobolev on Γ. we have $M_fT_{g_1,g_2,j}\in\mathcal{L}_p$ iff $g\in L_p$(specify the weight).

having this theorem, the assertion would be much easier. on the other hand, it is much easier to prove this theorem instead of the lower bound in Theorem 1.2 because now there are no commutators!!! and, of course, integral kernel of $T_{g_1,g_2,j}$ is positive. so, one can take $e_Q = f_Q = w(Q)^{-\frac{1}{2}}\chi_Q$.

[Zhijie will add some Comment about the idea in this section later]

Lemma 5.1. *Assume that P and Q are two w-measurable sets on $\mathbb{R}^N$ such that $P\subset Q$. Suppose that there exist constants $1<A<B<\infty$ such that*

$$Aw(P)\le w(Q)\le Bw(P). \tag{5.1}$$

There is a w-measurable function $H_{P,Q}$ on $\mathbb{R}^N$ satisfying

1. $|H_{P,Q}(x)|\le w(Q)^{-1/2}\chi_Q(x)$;
2. $\int_Q H_{P,Q}(x)dw(x)=0$;
3. $|(b)_P-(b)_Q|\approx w(Q)^{-1/2}|\langle b, H_{P,Q}\rangle_{L^2(\mathbb{R}^N,dw)}|$, *where the implicit constants depending only on A and B.*

Proof. Denote by $c_{A,B} := \min\left\{(A-1)^{1/2}, (B-1)^{-1/2}\right\}$ and let

$$H_{P,Q} := c_{A,B}\left(\frac{w(Q\backslash P)^{1/2}}{w(P)^{1/2}w(Q)^{1/2}}\chi_P(x) - \frac{w(P)^{1/2}}{w(Q)^{1/2}w(Q\backslash P)^{1/2}}\chi_{Q\backslash P}(x)\right).$$

Now we verify the function $H_{P,Q}$ satisfies all the required properties.

• **Verification of (1).**

It is obvious that $\operatorname{supp} H_{P,Q} \subset Q$. Next, we apply inequality (5.1) to see that

$$(A-1)w(P) \le w(Q\backslash P) = w(Q) - w(P) \le (B-1)w(P). \tag{5.2}$$

Thus, $c_{A,B}w(Q\backslash P)^{1/2} \le w(P)^{1/2}$ and $c_{A,B}w(P)^{1/2} \le w(Q\backslash P)^{1/2}$. This verifies property (1).

• **Verification of (2).**

Note that

$$\begin{aligned}\int_Q H_{P,Q}(x)dw(x) &= c_{A,B}\int_Q\left(\frac{w(Q\backslash P)^{1/2}}{w(P)^{1/2}w(Q)^{1/2}}\chi_P(x) - \frac{w(P)^{1/2}}{w(Q)^{1/2}w(Q\backslash P)^{1/2}}\chi_{Q\backslash P}(x)\right)dw(x)\\ &= c_{A,B}\left(\frac{w(Q\backslash P)^{1/2}w(P)^{1/2}}{w(Q)^{1/2}} - \frac{w(P)^{1/2}w(Q\backslash P)^{1/2}}{w(Q)^{1/2}}\right)\\ &= 0.\end{aligned}$$

This verifies property (2).

• **Verification of (3).**

We first show a stronger equality, that is,

$$c_{A,B}\big((b)_P - (b)_Q\big) = \langle b, H_{P,Q}\rangle_{L^2(\mathbb{R}^N,dw)}\frac{w(Q\backslash P)^{1/2}}{w(P)^{1/2}w(Q)^{1/2}}. \tag{5.3}$$

Indeed, we have

RHS of (5.3)

$$\begin{aligned}&= c_{A,B}\left(\frac{w(Q)-w(P)}{w(P)w(Q)}\int_P b(x)dw(x) - \frac{1}{w(Q)}\int_{Q\backslash P} b(x)dw(x)\right)\\ &= c_{A,B}\left(\fint_P b(x)dw(x) - \frac{1}{w(Q)}\int_P b(x)dw(x) - \left(\fint_Q b(x)dw(x) - \frac{1}{w(Q)}\int_P b(x)dw(x)\right)\right)\\ &= \text{LHS of (5.3)}.\end{aligned}$$

Combining (5.3) with (5.2), we deduce property (3) and then, Lemma 5.1. □

Lemma 5.2. *Let b and H be two w-measurable complex-valued functions on $\mathbb{R}^N$. Assume that there is a cube $Q \in \bigcup_{\nu=1}^{2^N}\mathbb{D}^\nu$ such that*

(1) $|H(x)| \le w(Q)^{-1/2}\chi_Q$ *for all* $x \in \mathbb{R}^N$;

(2) $\int_Q H(x)dw(x) = 0$.

There exist function sequences $\{e_{Q,s}\}_{s=1}^4$ and $\{f_{Q,s}\}_{s=1}^4$ such that

$$|e_{Q,s}| + |f_{Q,s}| \le w(Q)^{-1/2}\chi_{6\sqrt{N}Q},$$

and that

$$|\langle b, H\rangle_{L^2(\mathbb{R}^N,dw)}| \lesssim w(Q)^{1/2}\sum_{s=1}^{4}|\langle [b, R_{\mathrm{Dunkl},j}]e_{Q,s}, f_{Q,s}\rangle_{L^2(\mathbb{R}^N,dw)}|. \tag{5.4}$$

THE MOST IMPORTANT TASK FOR ZHIJIE. please get rid of complex median method in this proof. our functions are WITHOUT LOSS OF GENERALITY real-valued.

Proof. By Lemma 2.3, for any $k \in \mathbb{Z}$ and $Q \in \mathbb{D}_k^\nu$, since $Q \subset B(c_Q, \sqrt{N}2^{-k-1}) =: B_Q$, there exists a ball $\widehat{B_Q} := B(c_{\widehat{B_Q}}, \sqrt{N}2^{-k-1})$ such that $\|c_Q - c_{\widehat{B_Q}}\| = 5\sqrt{N}2^{-k-1}$, and that for every $(x,y) \in Q \times \widehat{B_Q}$, $K_{\mathrm{Dunkl},j}(x,y)$ does not change sign and satisfies

$$|K_{\mathrm{Dunkl},j}(x,y)| \gtrsim \frac{1}{w(B_Q)}. \tag{5.5}$$

It follows from the doubling inequality (2.2) that

$$w(B_Q) \sim w(\widehat{B_Q}) \sim w(Q).$$

By [17, Theorem 1.5], there exist $\theta \in [0, 2\pi)$, depending on $\widehat{B_Q}$ and b, and $\alpha_{\widehat{B_Q}}(b) \in \mathbb{C}$ such that if we denote

$$F_s^Q := \left\{x \in \widehat{B_Q} : \ -\frac{\pi}{4} + \frac{(s-1)\pi}{2} \le \arg\left(e^{i\theta}\big(\alpha_{\widehat{B_Q}}(b) - b(x)\big)\right) \le -\frac{\pi}{4} + \frac{s\pi}{2} \ \textit{or}\ b(x) = \alpha_{\widehat{B_Q}}(b)\right\},$$

where $s \in \{1,2,3,4\}$, and $\arg(z)$ is the argument of a complex number z, then $w(F_s^Q) \ge \frac{1}{16}w(\widehat{B_Q})$, and $F_1^Q \cup F_2^Q \cup F_3^Q \cup F_4^Q = \widehat{B_Q}$. $\alpha_{\widehat{B_Q}}(b)$ is called a complex median value of b over $\widehat{B_Q}$. Similarly, if we denote

$$E_s^Q := \left\{x \in Q : \ -\frac{\pi}{4} + \frac{(s-1)\pi}{2} \le \arg\left(e^{i\theta}\big(b(x) - \alpha_{\widehat{B_Q}}(b)\big)\right) \le -\frac{\pi}{4} + \frac{s\pi}{2} \ \textit{or}\ b(x) = \alpha_{\widehat{B_Q}}(b)\right\} \tag{5.6}$$

then $E_1^Q \cup E_2^Q \cup E_3^Q \cup E_4^Q = Q$.

It follows from the assumption of H that

$$\begin{aligned}
|\langle b, H\rangle_{L^2(\mathbb{R}^N,dw)}| &= \left|\int_Q b(x)H(x)dw(x)\right| \\
&= \left|\int_Q \big(b(x) - \alpha_{\widehat{B_Q}}(b)\big)H(x)dw(x)\right| \\
&\lesssim w(Q)^{-\frac{1}{2}}\int_Q |b(x) - \alpha_{\widehat{B_Q}}(b)|dw(x) \\
&\lesssim w(Q)^{-\frac{1}{2}}\sum_{s=1}^{4}\int_{E_s^Q}|b(x) - \alpha_{\widehat{B_Q}}(b)|dw(x) =: \sum_{s=1}^{4} M_s^Q.
\end{aligned}$$

Note that for any $s \in \{1,2,3,4\}$, if $x \in E_s^Q$ and $y \in F_s^Q$, then

$$\begin{aligned}
|b(x) - \alpha_{\widehat{B_Q}}(b)| &\le |e^{i\theta}(b(x) - \alpha_{\widehat{B_Q}}(b))| + |e^{i\theta}(\alpha_{\widehat{B_Q}}(b) - b(y))| \\
&\le 2|e^{i\theta}(b(x) - b(y))| = 2|b(x) - b(y)|.
\end{aligned} \tag{5.7}$$

Combining (5.5) with (5.7), we deduce that

$$\begin{aligned} M_s^Q &\lesssim w(Q)^{-\frac{1}{2}} \int_{E_s^Q} |b(x) - \alpha_{\widehat{B_Q}}(b)| dw(x) \cdot \frac{w(F_s^Q)}{w(Q)} \\ &\lesssim w(Q)^{-\frac{1}{2}} \int_{E_s^Q} \int_{F_s^Q} |b(x) - \alpha_{\widehat{B_Q}}(b)||K_{\mathrm{Dunkl},j}(x,y)| dw(y) dw(x) \\ &\lesssim w(Q)^{-\frac{1}{2}} \int_{E_s^Q} \int_{F_s^Q} |b(x) - b(y)||K_{\mathrm{Dunkl},j}(x,y)| dw(y) dw(x). \end{aligned} \tag{5.8}$$

Moreover, note that for any $x \in E_1^Q$ and $y \in F_1^Q$, the arguments of $e^{i\theta}(b(x) - \alpha_{\widehat{B_Q}}(b))$ and $e^{i\theta}(\alpha_{\widehat{B_Q}}(b) - b(y))$ both belong to $[-\frac{\pi}{4}, \frac{\pi}{4}]$. Thus,

$$-\frac{\pi}{4} \le \arg\Big(e^{i\theta}\big(b(x) - b(y)\big)\Big) \le \frac{\pi}{4}.$$

This implies that

$$|b(x) - b(y)| = \Big|e^{i\theta}\big(b(x) - b(y)\big)\Big| \le 2\mathrm{Re}\Big(e^{i\theta}\big(b(x) - b(y)\big)\Big). \tag{5.9}$$

Therefore,

$$\begin{aligned} M_1^Q &\lesssim w(Q)^{-\frac{1}{2}} \int_{E_1^Q} \int_{F_1^Q} \mathrm{Re}\Big(e^{i\theta}\big(b(x) - b(y)\big)\Big) K_{\mathrm{Dunkl},j}(x,y) dw(y) dw(x) \\ &\lesssim w(Q)^{-\frac{1}{2}} \left| \int_{E_1^Q} \int_{F_1^Q} e^{i\theta}\big(b(x) - b(y)\big) K_{\mathrm{Dunkl},j}(x,y) dw(y) dw(x) \right| \\ &= w(Q)^{-\frac{1}{2}} \left| \int_{E_1^Q} \int_{F_1^Q} \big(b(x) - b(y)\big) K_{\mathrm{Dunkl},j}(x,y) dw(y) dw(x) \right|. \end{aligned}$$

The other three terms M_2^Q, M_3^Q, M_4^Q can be handled in the same way as M_1^Q by rotation. Hence, for $s \in \{1,2,3,4\}$,

$$\begin{aligned} M_s^Q &\lesssim w(Q)^{-\frac{1}{2}} \left| \int_{E_s^Q} \int_{F_s^Q} \big(b(x) - b(y)\big) K_{\mathrm{Dunkl},j}(x,y) dw(y) dw(x) \right| \\ &= w(Q)^{\frac{1}{2}} \left\langle [b, R_{\mathrm{Dunkl},j}] \frac{\chi_{F_s^Q}}{w(Q)^{1/2}}, \frac{\chi_{E_s^Q}}{w(Q)^{1/2}} \right\rangle_{L^2(\mathbb{R}^N, dw)}. \end{aligned} \tag{5.10}$$

Letting

$$e_{Q,s} := \frac{\chi_{F_s^Q}}{w(Q)^{1/2}} \text{ and } f_{Q,s} := \frac{\chi_{E_s^Q}}{w(Q)^{1/2}},$$

which deduces (5.4). Clearly, $\operatorname{supp} E_s^Q \subset Q$. Moreover, since $\|c_Q - c_{\widehat{B_Q}}\| = 5\sqrt{N}2^{-k-1}$, we see that $\operatorname{supp} F_s^Q \subset 6\sqrt{N}Q$. This completes the proof of Lemma 5.2. □

[Zhijie may move the following definition to somewhere when writing the introduction]

Lemma 5.3. *For any $1 \le p < \infty$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$, we have*

$$\|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} \lesssim \|b\|_{B^p(\mathbb{R}^N, dw)} + \|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)}, \tag{5.11}$$

where we write

$$\|b\|_{B^p(\mathbb{R}^N,dw)} := \left(\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \frac{|b(x)-b(y)|^p}{w(B(x,\|x-y\|))^2}dw(x)dw(y)\right)^{1/p}$$

and

$$\|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)} := \Bigg(\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\times$$
$$\times \fint_{B(c_Q,A_2 2^{-k+1})}\fint_{B(\widetilde{\sigma}(\sigma(c_Q)),A_2 2^{-k+1})}|b(x)-b(y)|^p dw(x)dw(y)\Bigg)^{1/p}.$$

Proof. Let Γ be a fixed Weyl chamber. By decomposing

$$\mathbb{R}^N = \bigcup_{\sigma\in G}\sigma(\Gamma),$$

we write

$$\begin{aligned}
\|b\|^p_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} &= \sum_{\sigma,\widetilde{\sigma}\in G}\int_{\sigma(\Gamma)}\int_{\widetilde{\sigma}(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&= \sum_{\sigma\in G}\int_{\sigma(\Gamma)}\int_{\sigma(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{\widetilde{\sigma}(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&= \sum_{\sigma\in G}\int_{\sigma(\Gamma)}\int_{\sigma(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{D(y)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{E(y)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y),
\end{aligned}\tag{5.12}$$

where we denote by

$$D(y) := \left\{x\in\widetilde{\sigma}(\Gamma):\ \|x-y\|<2^{10}d(x,y)\right\},$$

$$E(y) := \left\{x\in\widetilde{\sigma}(\Gamma):\ \|x-y\|\geq 2^{10}d(x,y)\right\}.$$

It follows from (2.4) that $d(x,y)=\|x-\widetilde{\sigma}(\sigma(y))\|$ if $x\in\widetilde{\sigma}(\Gamma)$ and $y\in\sigma(\Gamma)$ and that $d(x,y)\leq\|x-y\|$ for all $x,y\in\mathbb{R}^N$. In particular, to estimate the first term on the right-hand side of (5.12), we note that for each $\sigma\in G$ and for every $x,y\in\sigma(\Gamma)$, $d(x,y)=\|x-y\|$. To estimate the second term, we apply the doubling inequality (2.2) to replace $d(x,y)$ by $\|x-y\|$. Consequently, the first two terms are dominated by $C\|b\|_{B^p(\mathbb{R}^N,dw)}$, where we used the facts that G is a finite group and w is a doubling measure. Thus, it remains to estimate the third term.

Let $\mathbb{D}_\Gamma := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_{\Gamma,k}$ be a system of dyadic cubes over Γ, which is constructed in Corollary 3.7. It follows that $\sigma(\mathbb{D}_\Gamma) := \bigcup_{k\in\mathbb{Z}} \sigma(\mathbb{D}_{\Gamma,k})$ is a system of dyadic cubes over $\sigma(\Gamma)$. Recall that A_2 is the constant given in Definition 2.4. We have

$$\begin{aligned}
&\int_{\sigma(\Gamma)}\int_{E(y)} \left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2} dw(x)dw(y)\\
&= \int_{\sigma(\Gamma)}\int_{E(y)} \left(\frac{\|x-\widetilde{\sigma}(\sigma(y))\|}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,\|x-\widetilde{\sigma}(\sigma(y))\|))^2} dw(x)dw(y)\\
&\le \sum_{k\in\mathbb{Z}} \int_{\sigma(\Gamma)}\int_{E_k(y)} \left(\frac{A_2 2^{-k}}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,A_2 2^{-k-1}))^2} dw(x)dw(y)\\
&= \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \int_Q\int_{E_k(y)} \left(\frac{A_2 2^{-k}}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,A_2 2^{-k-1}))^2} dw(x)dw(y), \qquad (5.13)
\end{aligned}$$

where

$$E_k(y) := \left\{x\in E(y):\ A_2 2^{-k-1} \le \|x-\widetilde{\sigma}(\sigma(y))\| \le A_2 2^{-k}\right\}.$$

It follows from the doubling inequality (2.2) and the G-invariance of the Dunkl measure that for every $y\in Q$ and $x\in E_k(y)\subset B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1})$ with $Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$,

$$w(B(x,A_2 2^{-k-1})) \sim w(B(\widetilde{\sigma}(\sigma(c_Q)),2^{-k})) \sim w(B(c_Q,2^{-k})) \sim w(Q).$$

Substituting this inequality into (5.13), we deduce that

$$\begin{aligned}
&\text{RHS of (5.13)}\\
&\lesssim \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \frac{2^{-kp}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_k(y)} \left(\frac{|b(x)-b(y)|}{\|x-y\|}\right)^p dw(x)dw(y)\\
&\lesssim \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\in\mathbb{Z}} \frac{2^{-kp}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}(y)} \left(\frac{|b(x)-b(y)|}{\|x-y\|}\right)^p dw(x)dw(y)\\
&\lesssim \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\in\mathbb{Z}} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}(y)} |b(x)-b(y)|^p dw(x)dw(y), \qquad (5.14)
\end{aligned}$$

where

$$E_{k,\ell}(y) := \left\{x\in E_k(y):\ A_2 2^{-\ell-1} \le \|x-y\| \le A_2 2^{-\ell}\right\}.$$

Note that for every $y\in\sigma(\Gamma)$ and $x\in E_{k,\ell}(y)$,

$$A_2 2^{-k-1} \le \|x-\widetilde{\sigma}(\sigma(y))\| = d(x,y) \le 2^{-10}\|x-y\| \le A_2 2^{-\ell-10}.$$

This implies that the summands on the right-hand side of (5.14) vanish provided $\ell > k-9$. To continue, for every $Q\in\mathbb{D}_\Gamma$, we denote by

$$U_k(Q) := B(c_Q, A_2 2^{-k+1}) \backslash B(c_Q, A_2 2^{-k-2}).$$

Note that the reflection σ is linear and preserves the Euclidean norm. Therefore, if $y\in Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$ and $x\in E_{k,\ell}(y)$, then

$$\|x-\widetilde{\sigma}(\sigma(c_Q))\| \le \|x-\widetilde{\sigma}(\sigma(y))\| + \|\widetilde{\sigma}(\sigma(y))-\widetilde{\sigma}(\sigma(c_Q))\| \le A_2 2^{-k} + A_2 2^{-(k+10)} \le A_2 2^{-(k+1)},$$

$$\|x-\widetilde{\sigma}(\sigma(c_Q))\| \ge \|x-\widetilde{\sigma}(\sigma(y))\| - \|\widetilde{\sigma}(\sigma(y))-\widetilde{\sigma}(\sigma(c_Q))\| \ge A_2 2^{-k-1} - A_2 2^{-(k+10)} \ge A_2 2^{-(k-2)},$$

$$\|x-c_Q\| \le \|x-y\| + \|y-c_Q\| \le A_2 2^{-\ell} + A_2 2^{-(k+10)} \le A_2 2^{-\ell+1},$$

$$\|x-c_Q\| \geq \|x-y\| - \|y-c_Q\| \geq A_2 2^{-\ell-1} - A_2 2^{-(k+10)} \geq A_2 2^{-\ell-2}.$$

In other words, we have $x \in U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q)$. Combining these facts, we see that

$$\begin{aligned}
&\text{RHS of (5.14)}\\
&= \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\leq k-9} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}(y)} |b(x)-b(y)|^p dw(x)dw(y)\\
&\lesssim \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\leq k-9} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y),
\end{aligned} \tag{5.15}$$

where

$$E_{k,\ell} := \widetilde{\sigma}(\Gamma) \cap U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q).$$

We *claim* that if $\ell \geq -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3$, then

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \cap B(c_Q, A_2 2^{-\ell+1}) = \varnothing. \tag{5.16}$$

In particular, we have $U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q) = \varnothing$. If this is not the case, then there exists an element $x_0 \in B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \cap B(c_Q, A_2 2^{-\ell+1})$. Equivalently, we have $\|x_0 - \widetilde{\sigma}(\sigma(c_Q))\| \leq A_2 2^{-k+1}$ and $\|x_0 - c_Q\| \leq A_2 2^{-\ell+1}$. Together with the inequality $\ell \leq k-9$, we see that

$$\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \leq \|x_0 - \widetilde{\sigma}(\sigma(c_Q))\| + \|x_0 - c_Q\| \leq A_2 2^{-k+1} + A_2 2^{-\ell+1} \leq A_2 2^{-\ell+2}.$$

It follows from the assumption $\ell \geq -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3$ that

$$A_2 2^{-\ell+2} \leq \frac{1}{2}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|.$$

Combining with the above two inequalities, we deduce that $c_Q = \widetilde{\sigma}(\sigma(c_Q)) \in \sigma(\Gamma) \cap \widetilde{\sigma}(\Gamma)$, which is a contradiction since each c_Q does not lie in the boundaries of the Weyl chambers. This ends the proof of (5.16).

Next, we *claim* that if $\ell \leq \min\left\{k-4, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) - 3\right\}$, then

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \subset B(c_Q, A_2 2^{-\ell-2}). \tag{5.17}$$

In particular, we have $U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q) = \varnothing$. Indeed, we take x to be an arbitrary element of $B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1})$, then

$$\|x-c_Q\| \leq \|x-\widetilde{\sigma}(\sigma(c_Q))\| + \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \leq A_2 2^{-k+1} + \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \leq A_2 2^{-\ell-2},$$

where the last inequality follows from the assumption. This ends the proof of (5.17).

Denote by

$$\ell_{k,Q} := \min\left\{k-4, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) - 3\right\},$$

$$L_{k,Q} := \min\left\{k-9, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3\right\}.$$

Combining (5.16) with (5.17), we see that

RHS of (5.15)

$$
\begin{aligned}
&= \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell_{k,Q}<\ell\le L_{k,Q}} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y) \\
&= \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} \sum_{-3<\ell+\log_2(A_2^{-1}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|)\le 4} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))}\times \\
&\qquad\qquad \times \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y). \qquad (5.18)
\end{aligned}
$$

Next, using the trivial inclusions $E_{k,\ell}\subset B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1})$ and $Q\subset B(c_Q, A_2 2^{-k+1})$, the doubling inequality (2.2) and the G-invariance of the Dunkl measure, we conclude that the right-hand side of (5.18) is dominated by

$$
C\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{B(c_Q,A_2 2^{-k+1})} \fint_{B(\widetilde{\sigma}(\sigma(c_Q)),A_2 2^{-k+1})} |b(x)-b(y)|^p dw(x)dw(y).
$$

This ends the proof of Lemma 5.3. □

With the collection $\{\mathbb{D}^\nu\}_{\nu=1}^{2^N}$ of dyadic cubes systems given in Lemma 2.7, we define the conditional expectation of a locally integrable function f on $\mathbb{R}^N$ by the expression

$$
E_k^\nu(f)(x) := \sum_{Q\in\mathbb{D}_k^\nu} (f)_Q \chi_Q(x),\ x\in\mathbb{R}^N.
$$

Moreover, we define the martingale difference of f by

$$
D_k^\nu(f) := E_{k+1}^\nu(f) - E_k^\nu(f).
$$

Definition 5.4. *For any $1\le p<\infty$ and $f\in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$, we define the martingale Besov (semi)norm of f by*

$$
\|f\|_{MB^p(\mathbb{R}^N,dw)} := \sum_{\nu=1}^{2^N} \Big(\sum_{k\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|D_k^\nu f(x)|^p}{w(B(x,2^{-k}))} dw(x) \Big)^{1/p}.
$$

Lemma 5.5. *Suppose that $1<p<\infty$ and $b\in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. We have*

$$
\|b\|_{B^p(\mathbb{R}^N,dw)} \lesssim \|b\|_{MB^p(\mathbb{R}^N,dw)}.
$$

Proof. We first decompose $\mathbb{R}^N\times\mathbb{R}^N$ into dyadic shells as follows.

$$
\begin{aligned}
\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \frac{|b(x)-b(y)|^p}{w(B(x,\|x-y\|))^2} dw(x)dw(y) &\le \sum_{k\in\mathbb{Z}} \iint_{\substack{2^{-k-1}<\|x-y\|\le 2^{-k} \\ x,y\in\mathbb{R}^N}} \frac{|b(x)-b(y)|^p}{w(B(x,\|x-y\|))^2} dw(x)dw(y) \\
&\le \sum_{k\in\mathbb{Z}} \iint_{\substack{2^{-k-1}<\|x-y\|\le 2^{-k} \\ x,y\in\mathbb{R}^N}} \frac{|b(x)-b(y)|^p}{w(B(x,2^{-k-1}))^2} dw(x)dw(y). \qquad (5.19)
\end{aligned}
$$

Fix $k\in\mathbb{Z}$ and let $\mathbb{D}_k^0$ be the standard system of dyadic cubes over $\mathbb{R}^N$ at scale 2^{-k} (see Notation 2.6). For each $Q\in\mathbb{D}_k^0$, one has

$$
I_Q := \big\{ y\in\mathbb{R}^N : \mathrm{dist}(y,Q)\le 2^{-k} \big\} \subset B(c_Q, \sqrt{N}2^{-k+1}),
$$

where we used the notation $\operatorname{dist}(y,Q)$ to denote the Euclidean distance from a point $y \in \mathbb{R}^N$ to a set Q. By Lemma 2.7, there exists $Q' \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu_{k-k_0}$, where $k_0 := \lfloor \log_2(24\sqrt{N}) \rfloor$, such that

$$B(c_Q, \sqrt{N}2^{-k+1}) \subset Q' \subset B(c_Q, 12N \cdot 2^{-k+1}).$$

With these observations, we conclude that

$$\begin{aligned}
&\text{RHS of (5.19)}\\
&\le \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^0_k} \int_Q \int_{I_Q} \frac{|b(x)-b(y)|^p}{w(B(x,2^{-k-1}))^2} dw(x)dw(y)\\
&\le \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^0_k} \int_{Q'} \int_{Q'} \frac{|b(x)-b(y)|^p}{w(B(x,2^{-k-1}))^2} dw(x)dw(y)\\
&\le \sum_{\nu=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\nu_{k-k_0}} \int_Q \int_Q \frac{|b(x)-b(y)|^p}{w(B(x,2^{-k-1}))^2} dw(x)dw(y), \qquad (5.20)
\end{aligned}$$

where in the last inequality we used the fact that each $Q' \in \mathbb{D}^\nu_{k-k_0}$ contains at most c (an absolute constant) $Q \in \mathbb{D}^0_k$. By the doubling inequality (2.2), for any $x, y \in Q \in \mathbb{D}^\nu_{k-k_0}$,

$$w(Q) \sim w(B(x,2^{-k-1})) \sim w(B(y,2^{-k-1})).$$

Therefore, by inserting and subtracting the conditional expectation $E^\nu_{k-k_0}(b)$, we deduce that

$$\begin{aligned}
&\text{RHS of (5.20)}\\
&\lesssim \sum_{\nu=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\nu_{k-k_0}} \int_Q \int_Q \frac{|b(x)-E^\nu_{k-k_0}(b)(x)|^p}{w(B(x,2^{-k-1}))^2} + \frac{|b(y)-E^\nu_{k-k_0}(b)(y)|^p}{w(B(x,2^{-k-1}))^2} dw(x)dw(y)\\
&\lesssim \sum_{\nu=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\nu_{k-k_0}} \int_Q \frac{|b(x)-E^\nu_{k-k_0}(b)(x)|^p}{w(B(x,2^{-k-1}))} dw(x)\\
&\lesssim \sum_{\nu=1}^{2^N} \sum_{k\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|b(x)-E^\nu_k(b)(x)|^p}{w(B(x,2^{-k}))} dw(x), \qquad (5.21)
\end{aligned}$$

where the last inequality used the change of index and the doubling inequality (2.2).

To proceed, we apply Lemma 2.8 to write

$$b(x) - E^\nu_k(b)(x) = \sum_{m=0}^{\infty} D^\nu_{k+m}(b)(x),$$

where the sum converges pointwise almost everywhere. Moreover, by the lower doubling inequality (2.2),

$$C^{-1}2^{mN} \le \frac{w(B(x,2^{-k}))}{w(B(x,2^{-(m+k)}))}.$$

Combining these facts, we deduce that

$$\left(\int_{\mathbb{R}^N} \frac{|b(x)-E_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))}\,dw(x)\right)^{1/p} \le \sum_{m\ge 0}\left(\int_{\mathbb{R}^N} \frac{|D_{k+m}^\nu(b)(x)|^p}{w(B(x,2^{-k}))}\,dw(x)\right)^{1/p}$$

$$\lesssim \sum_{m\ge 0} 2^{-mN/p}\left(\int_{\mathbb{R}^N} \frac{|D_{k+m}^\nu(b)(x)|^p}{w(B(x,2^{-(k+m)}))}\,dw(x)\right)^{1/p}. \tag{5.22}$$

Since $\sum_{m\ge 0} 2^{-mN/p} = (1-2^{-N/p})^{-1} < +\infty$, by discrete convolution Young's inequality, we have

$$\left(\sum_{k\in\mathbb{Z}}\int_{\mathbb{R}^N} \frac{|b(x)-E_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))}\,dw(x)\right)^{1/p} \lesssim \left(\sum_{k\in\mathbb{Z}}\int_{\mathbb{R}^N} \frac{|D_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))}\,dw(x)\right)^{1/p}.$$

Summing over ν yields the required assertion. □

Lemma 5.6. *Suppose that $1<p<\infty$ and $b\in L^1_{\mathrm{loc}}(\mathbb{R}^N,dw)$. If $[b,R_{\mathrm{Dunkl},j}]\in S^p$ for some $j\in\{1,2,\dots,N\}$, then*

$$\|b\|_{B^p(\mathbb{R}^N,dw)} \lesssim \|[b,R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Proof. By Lemma 5.5, it suffices to show that for each $\nu\in\{1,2,\dots,2^N\}$, we have

$$\left(\sum_{k\in\mathbb{Z}}\int_{\mathbb{R}^N} \frac{|D_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))}dw(x)\right)^{1/p} \lesssim \|[b,R_{\mathrm{Dunkl},j}]\|_{S^p}. \tag{5.23}$$

For any $Q\in\mathbb{D}_k^\nu$, we let $h_Q^1,h_Q^2,\dots,h_Q^{2^N-1}$ be a family of Haar functions as defined in Section 2.4. Next, we choose h_Q among these functions such that

$$h_Q := \left\{h_Q^i : \left|\int_Q b(x)h_Q^i(x)dw(x)\right| \text{ is maximal with respect to } i=1,2,\dots,2^N-1\right\}.$$

Note that for every $x\in Q$,

$$D_k^\nu(b)(x) = \sum_{i=1}^{2^N-1}\langle b,h_Q^i\rangle h_Q^i(x), \text{ with } \|h_Q^i\|_\infty \lesssim w(Q)^{-1/2}.$$

Hence, we have

$$\left(\fint_Q |D_k^\nu(b)(x)|^p dw(x)\right)^{1/p} \le C_{p,N}w(Q)^{-1/2}\left|\int_Q b(x)h_Q(x)dw(x)\right| \tag{5.24}$$

for some constant $C_{p,N}>0$. This, together with the doubling inequality (2.2), implies

$$\int_{\mathbb{R}^N} \frac{|D_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))}dw(x) \lesssim \sum_{Q\in\mathbb{D}_k^\nu}\fint_Q |D_k^\nu(b)(x)|^p dw(x)$$

$$\lesssim \sum_{Q\in\mathbb{D}_k^\nu} w(Q)^{-p/2}|\langle b,h_Q\rangle_{L^2(\mathbb{R}^N,dw)}|^p. \tag{5.25}$$

By Lemma 5.2, there exist NWO sequences $\{u_Q^{(s)}\}$ and $\{v_Q^{(s)}\}$, $s = 1, 2, 3, 4$, such that

$$|\langle b, h_Q\rangle_{L^2(\mathbb{R}^N,dw)}| \lesssim w(Q)^{1/2} \sum_{s=1}^{4} |\langle [b, R_{\mathrm{Dunkl},j}]u_Q^{(s)}, v_Q^{(s)}\rangle_{L^2(\mathbb{R}^N,dw)}|. \tag{5.26}$$

Substituting (5.26) into (5.25) and then applying Lemma 2.10, we deduce that

$$\begin{aligned}\sum_{k\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|D_k^\nu(b)(x)|^p}{w(B(x,2^{-k}))} dw(x) &\lesssim \sum_{s=1}^{4} \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}_k^\nu} |\langle [b, R_{\mathrm{Dunkl},j}]u_Q^{(s)}, v_Q^{(s)}\rangle_{L^2(\mathbb{R}^N,dw)}|^p \\ &\lesssim \|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}^p.\end{aligned}$$

This shows the inequality (5.23) and then finishes the proof of Lemma 5.6. □

Lemma 5.7. *Let T be a bounded linear operator on a Hilbert space H. For any $\xi, \eta \in H$, we have*

$$\langle T\xi, \eta\rangle \;=\; \mathrm{Tr}\big((\xi\otimes\eta)T\big),$$

where $\xi\otimes\eta$ denotes the rank-one operator defined by $(\xi\otimes\eta)(x) := \langle x, \eta\rangle\xi$ for $x \in H$.

Proof. For any $x \in H$, we have

$$(\xi\otimes\eta)T(x) = (\xi\otimes\eta)(Tx) = \langle Tx, \eta\rangle\xi = \langle x, T^*\eta\rangle\xi = (\xi\otimes T^*\eta)(x).$$

Hence,

$$(\xi\otimes\eta)T = \xi\otimes T^*\eta.$$

Since the trace of a rank-one operator $u\otimes v$ is given by $\mathrm{Tr}(u\otimes v) = \langle u, v\rangle$, we obtain

$$\mathrm{Tr}((\xi\otimes\eta)T) = \mathrm{Tr}(\xi\otimes T^*\eta) = \langle \xi, T^*\eta\rangle.$$

Finally, $\langle \xi, T^*\eta\rangle = \langle T\xi, \eta\rangle$, which completes the proof. □

Lemma 5.8. *Suppose that $N < p < \infty$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. Assume that $[b, R_{\mathrm{Dunkl},j}] \in S^p$ for some $j \in \{1, 2, \ldots, N\}$, then we have*

$$\|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)} \lesssim \|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Proof. Let $k_0 := \lfloor \log_2(24A_2)\rfloor$ be an integer such that $\frac{2^{-(k-k_0)-2}}{3} \le A_2 2^{-k+1} < \frac{2^{-(k-k_0)-1}}{3}$. By Lemma 2.7, for each $k \in \mathbb{Z}$ and $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$, there exist cubes $W_Q, P_Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}_{k-k_0}^\nu$ such that

$$B(c_Q, A_2 2^{-k+1}) \subset W_Q \subset B(c_Q, 12\sqrt{N}A_2 2^{-k+1}), \tag{5.27}$$

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \subset P_Q \subset B(\widetilde{\sigma}(\sigma(c_Q)), 12\sqrt{N}A_2 2^{-k+1}). \tag{5.28}$$

Together with the doubling inequality (2.2), we deduce that

$$
\begin{aligned}
&\|b\|^p_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)}\\
&\lesssim \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\fint_{W_Q}\fint_{P_Q}|b(x)-b(y)|^p dw(x)dw(y)\\
&=\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{\nu,\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^{\nu}_{k-k_0},W_Q\in\mathbb{D}^{\tau}_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\fint_{W_Q}\fint_{P_Q}|b(x)-b(y)|^p dw(x)dw(y)\\
&\lesssim\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ W_Q\in\mathbb{D}^{\tau}_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\fint_{W_Q}|b(x)-(b)_{W_Q}|^p dw(x)\\
&\qquad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{\nu=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^{\nu}_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\fint_{P_Q}|b(x)-(b)_{P_Q}|^p dw(x)\\
&\qquad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\sum_{\nu,\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^{\nu}_{k-k_0},W_Q\in\mathbb{D}^{\tau}_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q}-(b)_{W_Q}|^p\\
&=:\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}(\mathrm{I}^p_{\sigma,\tilde{\sigma}}+\mathrm{II}^p_{\sigma,\tilde{\sigma}}+\mathrm{III}^p_{\sigma,\tilde{\sigma}}).
\end{aligned}
$$

Since $Q\subset W_Q$ for all $Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$, we see that for each $\widetilde{Q}\in\mathbb{D}^{\tau}_{k-k_0}$, there are at most finite (an absolute constant) $Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$ such that $W_Q=\widetilde{Q}$. This implies that

$$
\begin{aligned}
\mathrm{I}^p_{\sigma,\tilde{\sigma}}&\lesssim\sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ W_Q\in\mathbb{D}^{\tau}_{k-k_0}}}\fint_{W_Q}|b(x)-(b)_{W_Q}|^p dw(x)\\
&\lesssim\sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\tau}_{k-k_0}}\fint_{Q}|b(x)-(b)_{Q}|^p dw(x)\\
&\lesssim\|b\|^p_{B^p(\mathbb{R}^N,dw)},
\end{aligned}
\tag{5.29}
$$

where in the last inequality we used the fact that oscillation space $\mathrm{OSC}^{p,p}(\mathbb{R}^N,dw)$ is equivalent to Besov space $B^p(\mathbb{R}^N,dw)$ (see e.g., [8, Proposition 17.5]). Similarly, the second term $\mathrm{II}_{\sigma,\tilde{\sigma}}$ is also bounded by $\|b\|_{B^p(\mathbb{R}^N,dw)}$. Together with Lemma 5.6, we deduce that

$$\mathrm{I}_{\sigma,\tilde{\sigma}}+\mathrm{II}_{\sigma,\tilde{\sigma}}\lesssim\|[b,R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Now we turn to the term $\mathrm{III}_{\sigma,\tilde{\sigma}}$. For each $Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^{\nu}_{k-k_0}$ and $k \in \mathbb{N}$, denote by $Q^{[k]}$ the kth strict dyadic ancestor. For convenience, we also set $Q^{[0]} := Q$. Pick $\alpha(Q) \in \mathbb{Z}$ such that $2^{\alpha(Q)} \le A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| < 2^{\alpha(Q)+1}$. Moreover, we let

$$s_Q := \log_2(2^{k+10+\alpha(Q)}) = k + 10 + \alpha(Q).$$

If $A_2 2^{-k+1} \le \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|$, then s_Q is a positive integer larger than 10. We have

$$\begin{aligned}
\mathrm{III}^p_{\sigma,\tilde{\sigma}} \lesssim & \sum_{\nu=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^{\nu}_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q} - (b)_{P_Q^{[s_Q]}}|^p \\
& + \sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ W_Q\in\mathbb{D}^{\tau}_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{W_Q} - (b)_{W_Q^{[s_Q]}}|^p \\
& + \sum_{\nu,\tau=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^{\nu}_{k-k_0}, W_Q\in\mathbb{D}^{\tau}_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[s_Q]}} - (b)_{W_Q^{[s_Q]}}|^p \\
=: & \ \mathrm{III}^p_{\sigma,\tilde{\sigma},1} + \mathrm{III}^p_{\sigma,\tilde{\sigma},2} + \mathrm{III}^p_{\sigma,\tilde{\sigma},3}.
\end{aligned}$$

By Lemma 5.1, for each $m \in \mathbb{N}$ and for each $Q \in \sigma(\mathbb{D}_\Gamma)$ with $P_Q \in \mathbb{D}^{\nu}_{k-k_0}$, there is a w-measurable function $H_{Q,m}$ on $\mathbb{R}^N$ (alternatively, $H_{Q,m}$ can be taken to be the Haar Basis element $h^1_{P_Q^{[m]}}$ (up to an absolute constant) if we enumerate $Ch(P_Q^{[m]}) = \{R_1, R_2, \dots, R_{2^N}\}$ so that $R_1 = P_Q^{[m-1]}$) satisfying

(1) $|H_{Q,m}(x)| \le w(P_Q^{[m]})^{-1/2}\chi_{P_Q^{[m]}}(x)$;

(2) $\int_{P_Q^{[m]}} H_{Q,m}(x)dw(x) = 0$;

(3) $|(b)_{P_Q^{[m-1]}} - (b)_{P_Q^{[m]}}| \approx w(P_Q^{[m]})^{-1/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}|$.

For simplicity, we denote by

$$\Lambda^\nu := \left\{(k,Q,m) : k\in\mathbb{Z}, Q\in\sigma(\mathbb{D}_{\Gamma,k+10}), A_2 2^{-k+1} \le \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|, P_Q\in\mathbb{D}^{\nu}_{k-k_0}, 1\le m\le s_Q\right\}.$$

By Minkowski's inequality,

$$\begin{aligned}
\mathrm{III}_{\sigma,\tilde{\sigma},1} &\lesssim \sum_{\nu=1}^{2^N}\left(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[m-1]}} - (b)_{P_Q^{[m]}}|^p\right)^{1/p} \\
&\lesssim \sum_{\nu=1}^{2^N}\left(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p} w(P_Q^{[m]})^{-p/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}|^p\right)^{1/p}.
\end{aligned} \tag{5.30}$$

By Lemma 5.2, there exist NWO sequences $\{e^{(s)}_{P_Q^{[m]}}\}$ and $\{f^{(s)}_{P_Q^{[m]}}\}$, $s=1,2,3,4$, such that

$$w(P_Q^{[m]})^{-1/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}| \lesssim \sum_{s=1}^{4} \big|\big\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\big\rangle_{L^2(\mathbb{R}^N,dw)}\big|.$$

Thus, the right-hand side of (5.30) is dominated by

$$\sum_{\nu=1}^{2^N}\sum_{s=1}^{4}\Bigg(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\big|\big\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\big\rangle_{L^2(\mathbb{R}^N,dw)}\big|^p\Bigg)^{1/p}.$$

Now taking $\{\varphi_{k,Q,m}\}_{(k,Q,m)\in\Lambda^\nu}$ with unit $\ell^{p'}$ norm, we apply the duality argument to estimate

$$\begin{aligned}
&\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\big\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\big\rangle_{L^2(\mathbb{R}^N,dw)}\varphi_{k,Q,m}\\
&=\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\mathrm{Tr}\Big(\big(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}}\big)[b, R_{\mathrm{Dunkl},j}]\Big)\varphi_{k,Q,m}\\
&=\mathrm{Tr}\Bigg(\Big(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\Big)\cdot[b, R_{\mathrm{Dunkl},j}]\Bigg)\\
&\le\Bigg\|\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\Bigg\|_{S^{p'}}\cdot\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p},
\end{aligned}\tag{5.31}$$

where the first equality follows from Lemma 5.7. It remains to show that

$$\sup_{\|\{\varphi_{k,Q,m}\}_{(k,Q,m)\in\Lambda^\nu}\|_{\ell^{p'}}=1}\Bigg\|\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\Bigg\|_{S^{p'}}\lesssim 1.\tag{5.32}$$

To show (5.32), we first write

$$\begin{aligned}
&\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\\
&=\sum_{m\ge1}\sum_{k\in\mathbb{Z}}\sum_{Q\in\Theta^\nu_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}}),
\end{aligned}\tag{5.33}$$

where

$$\Theta^\nu_{m,k}:=\big\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}): A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ \alpha(Q)\ge -k+m-10,\ P_Q\in\mathbb{D}^\nu_{k-k_0}\big\}.$$

Note that for any $Q\in\mathbb{D}^\nu_{k-k_0-m}$, there are at most $c2^{mN}$ cubes $W\in\sigma(\mathbb{D}_{\Gamma,k+10})$ such that $P_W^{[m]}=Q$, where c is an absolute constant. Denote these cubes by $Q^1_{[m]}$, $Q^2_{[m]}$, …, $Q^\iota_{[m]}$ with $\iota\le c2^{mN}$. Let $\delta_{Q,m,\ell}:=1$ if $P^{[m]}_{Q^\ell_{[m]}}=Q$ and $\delta_{Q,m,\ell}:=0$ otherwise. Let

$$\Xi^\nu_{m,k,\ell}:=\Big\{Q\in\mathbb{D}^\nu_{k-k_0-m}: A_2 2^{-k+1}\le\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|,\ \alpha(Q^\ell_{[m]})\ge -k+m-10\Big\}.$$

We have

$$
\begin{aligned}
&\text{RHS of (5.33)}\\
&= \sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^{\nu}_{m,k,\ell}} 2^{-k}\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|^{-1}\varphi_{k,Q^{\ell}_{[m]},m}\delta_{Q,m,\ell}(e^{(s)}_{Q}\otimes f^{(s)}_{Q})\\
&= 2^{-k_0}\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^{\nu}_{m,k+k_0+m,\ell}} 2^{-(k+m)}\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|^{-1}\varphi_{k+k_0+m,Q^{\ell}_{[m]},m}\delta_{Q,m,\ell}(e^{(s)}_{Q}\otimes f^{(s)}_{Q})\\
&= \sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\nu}_{k}}\lambda_Q(e^{(s)}_{Q}\otimes f^{(s)}_{Q}),
\end{aligned}
\tag{5.34}
$$

where for any $Q\in\mathbb{D}^{\nu}_{k}$,

$$
\begin{aligned}
\lambda_Q := 2^{-k_0}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} & 2^{-(k+m)}\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|^{-1}\varphi_{k+k_0+m,Q^{\ell}_{[m]},m}\delta_{Q,m,\ell}\times\\
&\times\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^{\ell}_{[m]})).
\end{aligned}
$$

It follows from equalities (5.33), (5.34) and Lemma 2.11 that

$$
\begin{aligned}
&\left\|\sum_{(k,Q,m)\in\Lambda^{\nu}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P^{[m]}_{Q}}\otimes f^{(s)}_{P^{[m]}_{Q}})\right\|_{S^{p'}}\\
&=\left\|\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\nu}_{k}}\lambda_Q(e^{(s)}_{Q}\otimes f^{(s)}_{Q})\right\|_{S^{p'}}\\
&\lesssim\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\nu}_{k}}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}.
\end{aligned}
\tag{5.35}
$$

By Hölder's inequality and the definition of λ_Q,

$$
\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\nu}_{k}}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}\\
&\lesssim\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^{\nu}_{k}}\left(\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|^{-p}\delta_{Q,m,\ell}\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^{\ell}_{[m]}))\right)^{\frac{p'}{p}}\times\right.\\
&\qquad\times\left(\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}\left|\varphi_{k+k_0+m,Q^{\ell}_{[m]},m}\delta_{Q,m,\ell}\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^{\ell}_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^{\ell}_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^{\ell}_{[m]}))\right|\right.
\end{aligned}
$$

Since $p > N$, we have

$$\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} 2^{-(k+m)p}\|c_{Q^\ell_{[m]}} - \widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-p}\delta_{Q,m,\ell}\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))$$
$$\lesssim \sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} 2^{-(k+m)p}\cdot 2^{kp} = C\sum_{m\geq 1} 2^{-m(p-N)} \lesssim 1.$$

Combining this with the same reindexing and counting argument used to derive (5.34), we obtain

$$\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\nu_k}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}\\
&\lesssim \Bigg(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\nu_k}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}\Big|\varphi_{k+k_0+m,Q^\ell_{[m]},m}\times\\
&\qquad\times\delta_{Q,m,\ell}\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^\ell_{[m]}} - \widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))\Big|^{p'}\Bigg)^{\frac{1}{p'}}\\
&= \left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k+k_0+m,\ell}}\left|\varphi_{k+k_0+m,Q^\ell_{[m]},m}\right|^{p'}\right)^{\frac{1}{p'}}\\
&= \left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k,\ell}}\left|\varphi_{k,Q^\ell_{[m]},m}\right|^{p'}\right)^{\frac{1}{p'}}\\
&= \left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{Q\in\Theta^\nu_{m,k}}|\varphi_{k,Q,m}|^{p'}\right)^{\frac{1}{p'}}\\
&= \left(\sum_{(k,Q,m)\in\Lambda^\nu}|\varphi_{k,Q,m}|^{p'}\right)^{\frac{1}{p'}}
\end{aligned}$$

(5.36)

$$= 1.$$

Substituting this inequality into (5.35), we obtain the desired inequality (5.32). Therefore, $\mathrm{III}_{\sigma,\tilde{\sigma},1}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. Similarly, $\mathrm{III}_{\sigma,\tilde{\sigma},2}$ is also dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$.

Next, we turn to the term III_3. Set $B_Q := B(c_Q, (2^{k_0+11} + 12\sqrt{N} + 1)\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|)$. We *claim* that

$$W_Q^{[s_Q]} \cup P_Q^{[s_Q]} \subset B_Q. \tag{5.37}$$

Indeed, we first note that for any $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$ with $A_2 2^{-k+1} \leq \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|$ and with $P_Q \in \mathbb{D}^\nu_{k-k_0}, W_Q \in \mathbb{D}^\tau_{k-k_0}$, we have

$$\ell(P_Q^{[s_Q]}) = \ell(W_Q^{[s_Q]}) = 2^{k_0+10+\alpha(Q)} \leq \frac{2^{k_0+10}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|}{A_2}. \tag{5.38}$$

Combining this inequality with the inclusions that $W_Q^{[s_Q]} \subseteq B(c_{W_Q^{[s_Q]}}, A_2\ell(W_Q^{[s_Q]}))$ and (5.27), we deduce that for any $x \in W_Q^{[s_Q]}$,

$$\begin{aligned}\|x - c_Q\| &\le \|x - c_{W_Q^{[s_Q]}}\| + \|c_{W_Q^{[s_Q]}} - c_{W_Q}\| + \|c_{W_Q} - c_Q\| \\ &\le 2A_2\ell(W_Q^{[s_Q]}) + 12\sqrt{N}A_2 2^{-k+1} \\ &\le (2^{k_0+11} + 12\sqrt{N})\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|.\end{aligned}$$

Similarly, combining (5.38) with the inclusions $P_Q^{[s_Q]} \subseteq B(c_{P_Q^{[s_Q]}}, A_2\ell(P_Q^{[s_Q]}))$ and (5.28), we deduce that for any $x \in P_Q^{[s_Q]}$,

$$\begin{aligned}\|x - c_Q\| &\le \|x - c_{P_Q^{[s_Q]}}\| + \|c_{P_Q^{[s_Q]}} - c_{P_Q}\| + \|c_{P_Q} - \widetilde{\sigma}(\sigma(c_Q))\| + \|\widetilde{\sigma}(\sigma(c_Q)) - c_Q\| \\ &\le 2A_2\ell(P_Q^{[s_Q]}) + 12\sqrt{N}A_2 2^{-k+1} + \|\widetilde{\sigma}(\sigma(c_Q)) - c_Q\| \\ &\le (2^{k_0+11} + 12\sqrt{N} + 1)\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|\end{aligned}$$

This finishes the proof of (5.37).

By Lemma 2.7, for any $Q \in \sigma(\mathbb{D}_\Gamma)$, there exists a cube $E_Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu$ such that

$$B_Q \subset E_Q \subset 12\sqrt{N}B_Q. \tag{5.39}$$

We have

$$\begin{aligned}\mathrm{III}^p_{\sigma,\tilde{\sigma},3} &\lesssim \sum_{\nu=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\| \\ P_Q\in\mathbb{D}^\nu_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[s_Q]}} - (b)_{E_Q}|^p \\ &\quad + \sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\| \\ W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{W_Q^{[s_Q]}} - (b)_{E_Q}|^p \\ &=: \mathrm{III}^p_{\sigma,\tilde{\sigma},31} + \mathrm{III}^p_{\sigma,\tilde{\sigma},32}.\end{aligned} \tag{5.40}$$

Next, we combine (5.39) with (5.37) and the doubling inequality (2.2) to see that

$$w(P_Q^{[s_Q]}) \sim w(E_Q).$$

By Lemma 5.1, for each $Q \in \sigma(\mathbb{D}_\Gamma)$ with $P_Q \in \mathbb{D}^\nu_{k-k_0}$, there is a w-measurable function $H_{P_Q^{[s_Q]},E_Q}$ on $\mathbb{R}^N$ (unlike the function $H_{Q,m}$ used in estimating $\mathrm{III}_{\sigma,\tilde{\sigma},1}$ and $\mathrm{III}_{\sigma,\tilde{\sigma},2}$, $H_{P_Q^{[s_Q]},E_Q}$ cannot be chosen from an arbitrary Haar basis) satisfying

(1) $|H_{P_Q^{[s_Q]},E_Q}(x)| \lesssim w(E_Q)^{-1/2}\chi_{E_Q}(x)$;

(2) $\int_{E_Q} H_{P_Q^{[s_Q]},E_Q}(x)dw(x) = 0$;

(3) $|(b)_{P_Q^{[s_Q]}} - (b)_{E_Q}| \approx w(E_Q)^{-1/2}|\langle b, H_{P_Q^{[s_Q]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}|$.

For simplicity, we denote by

$$\Upsilon^\nu := \left\{(k,Q) : k\in\mathbb{Z}, Q\in\sigma(\mathbb{D}_{\Gamma,k+10}), A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|, P_Q\in\mathbb{D}^\nu_{k-k_0}\right\}.$$

Therefore,

(5.41)
$$\mathrm{III}^p_{\sigma,\tilde{\sigma},31}\lesssim \sum_{\nu=1}^{2^N}\sum_{(k,Q)\in\Upsilon^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} w(E_Q)^{-p/2}|\langle b, H_{P_Q^{[sQ]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}|^p.$$

By Lemma 5.2, there exist NWO sequences $\{\xi^{(s)}_{E_Q}\}$ and $\{\eta^{(s)}_{E_Q}\}$, $s=1,2,3,4$, such that

$$w(E_Q)^{-1/2}|\langle b, H_{P_Q^{[sQ]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}|\lesssim \sum_{s=1}^4 \left|\left\langle [b,R_{\mathrm{Dunkl},j}]\xi^{(s)}_{E_Q},\eta^{(s)}_{E_Q}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|.$$

Substituting this inequality into (5.41) and then repeating the duality argument as in (5.31), we deduce that

$$\begin{aligned}\mathrm{III}_{\sigma,\tilde{\sigma},31}&\lesssim \sum_{s=1}^4\sum_{\nu=1}^{2^N}\left(\sum_{(k,Q)\in\Upsilon^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} w(E_Q)^{-p/2}\left|\left\langle [b,R_{\mathrm{Dunkl},j}]\xi^{(s)}_{E_Q},\eta^{(s)}_{E_Q}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|^p\right)^{1/p}\\ &\lesssim \sum_{s=1}^4\sum_{\nu=1}^{2^N}\sup_{\|\{\psi_{k,Q}\}_{(k,Q)\in\Upsilon^\nu}\|_{\ell^{p'}}=1}\left\|\sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\right\|_{S^{p'}}\|[b,R_{\mathrm{Dunkl},j}]\|_{S^p}.\end{aligned}\tag{5.42}$$

We *claim* that

(5.43)
$$\sup_{\|\{\psi_{k,Q}\}_{(k,Q)\in\Upsilon^\nu}\|_{\ell^{p'}}=1}\left\|\sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\right\|_{S^{p'}}\lesssim 1.$$

To show (5.43), we first write

$$\begin{aligned}&\sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\\ &=\sum_{m\ge1}\sum_{k\in\mathbb{Z}}\sum_{Q\in U^\nu_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\\ &=\sum_{\tau=1}^{2^N}\sum_{m\ge1}\sum_{k\in\mathbb{Z}}\sum_{Q\in U^{\nu,\tau}_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q}),\end{aligned}\tag{5.44}$$

where

$$U^\nu_{m,k}:=\left\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) : A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ \ell(E_Q)=2^{-(k-m+10)},\ P_Q\in\mathbb{D}^\nu_{k-k_0}\right\},$$

$$U^{\nu,\tau}_{m,k}:=\left\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) : A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ E_Q\in\mathbb{D}^\tau_{k-m+10},\ P_Q\in\mathbb{D}^\nu_{k-k_0}\right\}.$$

Note that for any $Q\in\mathbb{D}^\tau_{k-m+10}$, there are at most $c2^{mN}$ cubes $W\in\sigma(\mathbb{D}_{\Gamma,k+10})$ such that $E_W=Q$, where c is an absolute constant. Denote these cubes by

$Q^1_{\langle m\rangle}, Q^2_{\langle m\rangle}, \dots, Q^\varsigma_{\langle m\rangle}$ with $\varsigma \le c2^{mN}$. Let $\tilde\delta_{Q,m,\ell} = 1$ if $E_{Q^\ell_{\langle m\rangle}} = Q$ and $\tilde\delta := 0$ otherwise. Let

$$\Delta^{\nu,\tau}_{m,k,l} := \left\{ Q \in \mathbb{D}^\tau_{k-m+10} : A_2 2^{-k+1} \le \|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|,\ P_{Q^\ell_{\langle m\rangle}} \in \mathbb{D}^\nu_{k-k_0} \right\}.$$

We have

$$\begin{aligned}
&\text{RHS of (5.44)}\\
&= \sum_{\tau=1}^{2^N} \sum_{m\ge1} \sum_{k\in\mathbb{Z}} \sum_{\ell=1}^{c2^{mN}} \sum_{Q\in\Delta^{\nu,\tau}_{m,k,\ell}} 2^{-k} \|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1} \tilde\delta_{Q,m,\ell} \psi_{k,Q^\ell_{\langle m\rangle}} (\xi^{(s)}_Q \otimes \eta^{(s)}_Q)\\
&= 2^{10} \sum_{\tau=1}^{2^N} \sum_{m\ge1} \sum_{k\in\mathbb{Z}} \sum_{\ell=1}^{c2^{mN}} \sum_{Q\in\Delta^{\nu,\tau}_{m,k+m-10,\ell}} 2^{-(k+m)} \|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1} \tilde\delta_{Q,m,\ell} \psi_{k+m-10,Q^\ell_{\langle m\rangle}} (\xi^{(s)}_Q \otimes \eta^{(s)}_Q)\\
&= \sum_{\tau=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\tau_k} a^\nu_Q (\xi^{(s)}_Q \otimes \eta^{(s)}_Q),
\end{aligned} \tag{5.45}$$

where for any $Q \in \mathbb{D}^\tau_k$,

$$\begin{aligned}
a^\nu_Q := 2^{10} \sum_{m\ge1} \sum_{\ell=1}^{c2^{mN}} & 2^{-(k+m)} \|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1} \times\\
&\times \tilde\delta_{Q,m,\ell} \psi_{k+m-10,Q^\ell_{\langle m\rangle}} \chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|) \chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}}).
\end{aligned}$$

It follows from equalities (5.44), (5.45) and Lemma 2.11 that

$$\begin{aligned}
&\left\| \sum_{(k,Q)\in\Upsilon^\nu} 2^{-k} \|c_Q - \widetilde\sigma(\sigma(c_Q))\|^{-1} \psi_{k,Q} (\xi^{(s)}_{E_Q} \otimes \eta^{(s)}_{E_Q}) \right\|_{S^{p'}}\\
&\lesssim \sum_{\tau=1}^{2^N} \left\| \sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\tau_k} a^\nu_Q (\xi^{(s)}_Q \otimes \eta^{(s)}_Q) \right\|_{S^{p'}}\\
&\lesssim \sum_{\tau=1}^{2^N} \left(\sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\tau_k} |a^\nu_Q|^{p'} \right)^{\frac{1}{p'}}.
\end{aligned} \tag{5.46}$$

By Hölder's inequality and the definition of a^ν_Q,

$$\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\tau_k} |a^\nu_Q|^{p'} \right)^{\frac{1}{p'}}\\
&\lesssim \left(\sum_{k\in\mathbb{Z}} \sum_{Q\in\mathbb{D}^\tau_k} \left(\sum_{m\ge1} \sum_{\ell=1}^{c2^{mN}} 2^{-(k+m)p} \|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-p} \tilde\delta_{Q,m,\ell} \right)^{\frac{p'}{p}} \times\\
&\quad \times \left(\sum_{m\ge1} \sum_{\ell=1}^{c2^{mN}} |\tilde\delta_{Q,m,\ell} \psi_{k+m-10,Q^\ell_{\langle m\rangle}} \chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}} - \widetilde\sigma(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|) \chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}})|^{p'} \right) \right)^{\frac{1}{p'}}.
\end{aligned} \tag{5.47}$$

Observe that $\tilde{\delta}_{Q,m,\ell} \in \{0,1\}$ and that $\tilde{\delta}_{Q,m,\ell} = 1$ only when $\big\| c_{Q^\ell_{\langle m\rangle}} - \widetilde{\sigma}\big(\sigma(c_{Q^\ell_{\langle m\rangle}})\big)\big\| \geq A_2\, 2^{-k+1}$. This, in combination with the assumption $p > N$, implies

(5.48)
$$\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} 2^{-(k+m)p}\|c_{Q^\ell_{\langle m\rangle}} - \widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-p}\tilde{\delta}_{Q,m,\ell} \lesssim \sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} 2^{-(k+m)p}(2^{-k})^{-p} \lesssim \sum_{m\geq 1} 2^{-m(p-N)} \lesssim 1.$$

Substituting (5.48) into (5.47) and then applying a similar argument as in (5.36), we deduce that

$$\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k} |a^\nu_Q|^{p'}\right)^{\frac{1}{p'}} \\
&\lesssim \Bigg(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} |\psi_{k+m-10,Q^\ell_{\langle m\rangle}} \times \\
&\qquad \times \tilde{\delta}_{Q,m,\ell}\chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}} - \widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|)\chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}})|^{p'}\Bigg)^{\frac{1}{p'}} \\
&= \left(\sum_{(k,Q)\in\Upsilon^\nu} |\psi_{k,Q}|^{p'}\right)^{\frac{1}{p'}} \\
&\lesssim 1.
\end{aligned}$$

Substituting this inequality into (5.46), we obtain the inequality (5.43). Together with the inequality (5.42), we see that $\mathrm{III}_{\sigma,\tilde{\sigma},31}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. Similarly, $\mathrm{III}_{\sigma,\tilde{\sigma},32}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. This finishes the proof of Lemma 5.8. □

Proposition 5.9. *Suppose $N \geq 1$, $p > N$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. Assume that $[b, R_{\mathrm{Dunkl},j}] \in S^p$ for some $j \in \{1, 2, \ldots, N\}$. We have*

$$\|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} \lesssim \|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Proof. It follows from Lemma 5.3 that

$$\|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} \lesssim \|b\|_{B^p(\mathbb{R}^N,dw)} + \|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)}. \tag{5.49}$$

By Lemmas 5.6 and 5.8, each term is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. The completes the proof of Proposition 5.9. □

6. Poincaré inequality over Weyl chamber

Definition 6.1. *Let $U \subset \mathbb{R}^N$ be a convex set. We say that a non-negative function f is* log-concave on U *if for all $x, y \in U$ and all $t \in (0,1)$,*

$$f((1-t)x + ty) \;\geq\; f(x)^{1-t}\, f(y)^t.$$

Notation 6.2. *Define the projection $\mathrm{proj}_{\mathbb{R}^{N_0}} : \mathbb{R}^{N_0} \times \mathbb{R}^{N-N_0} \to \mathbb{R}^{N_0}$ by*

$$\mathrm{proj}_{\mathbb{R}^{N_0}}(u, z) := u.$$

Lemma 6.3. *Let ϕ be the function as in (3.5). The function $\xi \mapsto \phi(\mathrm{proj}_{\mathbb{R}^{N_0}}(\xi))$ is log-concave on Ω.*

Proof. Fix $a \in \mathbb{R}_+^{N_0} \setminus \{0\}$ and set $L(u) := a \cdot u$. On the open set $\{L > 0\}$,

$$\nabla \log L(u) = \frac{a}{a \cdot u}, \qquad \nabla^2 \log L(u) = -\frac{a\, a^{\mathsf{T}}}{(a \cdot u)^2},$$

which is negative semidefinite. Hence $\log L$ is concave on $\{L > 0\}$, and therefore L is log-concave on $\{L > 0\}$. Fix $u_1, u_2 \in \mathbb{R}_+^{N_0}$ and $t \in (0,1)$. For $\varepsilon > 0$ set

$$u_1^\varepsilon := u_1 + \varepsilon a, \qquad u_2^\varepsilon := u_2 + \varepsilon a.$$

Then $u_1^\varepsilon, u_2^\varepsilon \in \mathbb{R}_+^{N_0}$ and

$$L(u_i^\varepsilon) = a \cdot u_i + \varepsilon\, a \cdot a = L(u_i) + \varepsilon \|a\|^2 \ > \ 0, \qquad i = 1, 2,$$

so $u_1^\varepsilon, u_2^\varepsilon \in \{L > 0\}$. Since L is log-concave on $\{L > 0\}$, we have

$$L\big((1-t)u_1^\varepsilon + t u_2^\varepsilon\big) \ \geq \ L(u_1^\varepsilon)^{1-t}\, L(u_2^\varepsilon)^t.$$

Using the linearity of L,

$$L\big((1-t)u_1^\varepsilon + t u_2^\varepsilon\big) = L\big((1-t)u_1 + t u_2\big) + \varepsilon \|a\|^2, \qquad L(u_i^\varepsilon) = L(u_i) + \varepsilon \|a\|^2.$$

Letting $\varepsilon \to 0$ and using the continuity of L and of the map $s \mapsto s^\vartheta$ on $[0, \infty)$ for every $\vartheta > 0$, we obtain

$$L\big((1-t)u_1 + t u_2\big) \ \geq \ L(u_1)^{1-t}\, L(u_2)^t.$$

Thus L is log-concave on all of $\mathbb{R}_+^{N_0}$.

For each $v \in R_+$, we apply the above property with $a = a_v := (a_{v,1}, \ldots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}$ and $L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i = a_v \cdot u$ to conclude that each L_v is log-concave on $\mathbb{R}_+^{N_0}$. Since

$$\log \phi(u) = -\log |\det(\Phi)| + 2 \sum_{v \in R_+} k(v) \log L_v(u)$$

is the sum of a constant and a nonnegative linear combination of concave functions, ϕ is log-concave on $\mathbb{R}_+^{N_0}$.

Finally, note that for $\xi_1 = (u_1, z_1) \in \Omega$ and $\xi_2 = (u_2, z_2) \in \Omega$ with $u_1, u_2 \in \mathbb{R}_+^{N_0}$ and $z_1, z_2 \in \mathbb{R}^{N - N_0}$,

$$\phi(\mathrm{proj}_{\mathbb{R}^{N_0}}((1-t)\xi_1 + t\xi_2)) = \phi((1-t)u_1 + t u_2) \geq \phi(u_1)^{1-t} \phi(u_2)^t = \phi(\mathrm{proj}_{\mathbb{R}^{N_0}}(\xi_1))^{1-t} \phi(\mathrm{proj}_{\mathbb{R}^{N_0}}(\xi_2))^t.$$

so $\xi \mapsto \phi(\mathrm{proj}_{\mathbb{R}^{N_0}}(\xi))$ is log-concave on Ω. The proof of Lemma 6.3 is complete. □

Definition 6.4. *Let $f : \mathbb{R}^N \to [0, \infty)$ be Lebesgue measurable with $0 < \int_{\mathbb{R}^N} f(x) dx < \infty$. The* induced measure *$\pi_f$ is defined as the probability measure*

$$\pi_f(A) := \frac{\int_A f(x)\, dx}{\int_{\mathbb{R}^N} f(x)\, dx}, \quad \textit{for all measurable set } A \subset \mathbb{R}^N.$$

Proposition 6.5. [16, Theorem 4.2] *Let f be a log-concave function whose support has diameter D, and let π_f be the induced measure. Then for any partition of $\mathbb{R}^N$ into measurable sets S_1, S_2, S_3,*

$$\pi_f(S_3) \ \geq \ \frac{2\, \mathrm{dist}(S_1, S_2)}{D} \min\{\pi_f(S_1), \pi_f(S_2)\}.$$

Definition 6.6. *Let (X, d, v) be a metric measure space. For a Borel set $A \subset X$ and $\varepsilon > 0$, the closed ε-neighbourhood of A is defined as*

$$A_\varepsilon := \{x \in X : \ \mathrm{dist}(x, A) \le \varepsilon\}.$$

The codimension Minkowski content of $A \subset X$ is defined as

$$v^+(A) := \liminf_{\varepsilon \to 0} \frac{v(A_\varepsilon \backslash A)}{\varepsilon}.$$

The following lemma establishes a Cheeger-type inequality.

Lemma 6.7. *Let $K \subset \mathbb{R}^N$ be a convex body (i.e. a compact convex set with nonempty interior) with diameter $D := \mathrm{diam}(K)$. Let $\mathbb{P}$ be a Borel probability measure on K of the form $d\mathbb{P} = \rho\, dx$, where dx is N-dimensional Lebesgue measure and ρ is log-concave on K. Then for every measurable $A \subset K$,*

$$\mathbb{P}^+(A) \ \ge \ \frac{2}{D} \min\{\mathbb{P}(A),\, 1 - \mathbb{P}(A)\}.$$

Proof. Define $f : \mathbb{R}^N \to [0, \infty)$ by $f = \rho$ on K and $f = 0$ on $\mathbb{R}^N \setminus K$. Since K is convex and ρ is log-concave on K, f is log-concave on $\mathbb{R}^N$. Moreover $\mathrm{supp}(f) \subset \overline{K}$, hence $\mathrm{diam}(\mathrm{supp}(f)) \le D$. Since $\int_{\mathbb{R}^N} f\, dx = \int_K \rho\, dx = 1$, we have $\pi_f = \mathbb{P}$ (as measures on $\mathbb{R}^N$ supported on K). Fix $\varepsilon > 0$ and let $A_\varepsilon := \{x \in K : \ \mathrm{dist}(x, A) \le \varepsilon\}$. Consider the measurable partition of $\mathbb{R}^N$ given by

$$S_1 := A, \qquad S_2 := K \setminus A_\varepsilon, \qquad S_3 := (A_\varepsilon \setminus A) \cup (\mathbb{R}^N \setminus K).$$

Then $\mathrm{dist}(S_1, S_2) \ge \varepsilon$. Applying Proposition 6.5 to this partition and using $\mathrm{diam}(\mathrm{supp}(f)) \le D$ gives

$$\mathbb{P}(A_\varepsilon) - \mathbb{P}(A) = \pi_f(S_3) \ \ge \ \frac{2\varepsilon}{D} \min\{\pi_f(S_1), \pi_f(S_2)\} = \frac{2\varepsilon}{D} \min\{\mathbb{P}(A), 1 - \mathbb{P}(A_\varepsilon)\}.$$

Divide by ε:

$$\frac{\mathbb{P}(A_\varepsilon) - \mathbb{P}(A)}{\varepsilon} \ \ge \ \frac{2}{D} \min\{\mathbb{P}(A), 1 - \mathbb{P}(A_\varepsilon)\}.$$

Now as $\varepsilon \downarrow 0$, the sets A_ε decrease to $\overline{A} \cap K$. Hence, $\mathbb{P}(A_\varepsilon) \downarrow \mathbb{P}(\overline{A} \cap K)$. If $\mathbb{P}(\overline{A} \cap K) > \mathbb{P}(A)$ then $\mathbb{P}(A_\varepsilon) - \mathbb{P}(A) \ge c > 0$ for small ε, so $\mathbb{P}^+(A) = +\infty$ and the inequality is trivial. Otherwise $\mathbb{P}(\overline{A} \cap K) = \mathbb{P}(A)$, so $1 - \mathbb{P}(A_\varepsilon) \to 1 - \mathbb{P}(A)$. Taking $\liminf_{\varepsilon \to 0}$ yields the desired result. □

Definition 6.8. *Let $(K, \mathbb{P})$ be a probability space and $f \in L^1(\mathbb{P})$. We say that $m_f \in \mathbb{R}$ is a median value of f over K with respect to the measure $\mathbb{P}$ if*

$$\mathbb{P}(\{x \in K : f(x) \ge m_f\}) \ge \frac{1}{2}, \qquad \mathbb{P}(\{x \in K : f(x) \le m_f\}) \ge \frac{1}{2}.$$

A median value always exists, but may not be unique (see for example [11]). In what follows, we let m_f be an arbitrary median value of f.

Lemma 6.9. *Let $K \subset \mathbb{R}^N$ be a convex body (i.e. a compact convex set with nonempty interior) with diameter $D := \mathrm{diam}(K)$. Let $\mathbb{P}$ be a Borel probability measure on K of the form $d\mathbb{P} = \rho(x)\, dx$, where dx is N-dimensional Lebesgue measure and ρ is a log-concave function on K. Let $f \in \mathrm{Lip}(\overline{K})$ and let m_f be a median value of f. Then*

$$\int_K |f(x) - m_f|\, d\mathbb{P}(x) \ \le \ \frac{D}{2} \int_K |\nabla f(x)|\, d\mathbb{P}(x).$$

Proof. The proof is a direct consequence of the combination of Lemma 6.7 with [1, Theorem 1.1]. Indeed, we define the shift-invariant functional

$$L(h) := \inf_{a\in\mathbb{R}} \int_K |h(x) - a|\, d\mathbb{P}(x).$$

Note that for any measurable set $A \subset K$, we have $\max\{L(\mathbf{1}_A), L(-\mathbf{1}_A)\} = \min\{\mathbb{P}(A), 1-\mathbb{P}(A)\}$. This, together with Lemma 6.7, implies that the condition [1, (1.8)] holds. By [1, Theorem 1.1],

$$\int_K |\nabla f(x)|\, d\mathbb{P}(x) \;\geq\; \frac{2}{D} \inf_{a\in\mathbb{R}} \int_K |f(x) - a|\, d\mathbb{P}(x) = \frac{2}{D} \int_K |f(x) - m_f|\, d\mathbb{P}(x), \tag{6.1}$$

where the last equality follows from the definition of median value. □

Definition 6.10. *We say that $R' \subset \Omega$ is an axis–parallel rectangle if*

$$R' := I_1 \times \cdots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ supports a rectangle-version $(1,1)$–Poincaré inequality.

Lemma 6.11. *For every axis–parallel rectangle $R' \subset \Omega$ and every Lipschitz function $f : \Omega \to \mathbb{C}$, we have*

$$\fint_{R'} \fint_{R'} |f(x) - f(y)|\, d\mu(x)\, d\mu(y) \;\leq\; \operatorname{diam}(R') \fint_{R'} |\nabla f(x)|\, d\mu(x).$$

Proof. By decomposing $f = \operatorname{Re} f + i \operatorname{Im} f$ and then using triangular inequality, it suffices to show the conclusion for Lipschitz function $f : \Omega \to \mathbb{R}$.

Define a probability measure supported on R' by

$$d\mathbb{P}(x) := \frac{1}{\mu(R')}\, d\mu(x) = \frac{\phi(\operatorname{proj}_{\mathbb{R}^{N_0}}(x))}{\mu(R')}\, dx,$$

where the second equality follows from Lemma 3.9. It is equivalent to prove

$$\int_{R'} \int_{R'} |f(x) - f(y)|\, d\mathbb{P}(x)\, d\mathbb{P}(y) \;\leq\; \operatorname{diam}(R') \int_{R'} |\nabla f(x)|\, d\mathbb{P}(x). \tag{6.2}$$

To show the inequality (6.2), we first apply Lemma 6.3 to see that the function $\xi \mapsto \phi(\operatorname{proj}_{\mathbb{R}^{N_0}}(\xi))$ is log-concave on Ω. Since R' is convex and $R' \subset \Omega$, the function $x \mapsto \phi(\operatorname{proj}_{\mathbb{R}^{N_0}}(x))$ is log-concave on R'. By Lemma 6.9,

$$\int_{R'} \int_{R'} |f(x) - f(y)|\, d\mathbb{P}(x)\, d\mathbb{P}(y) \leq 2 \int_{R'} |f(x) - m_f|\, d\mathbb{P}(x) \leq \operatorname{diam}(R') \int_{R'} |\nabla f(x)|\, d\mathbb{P}(x),$$

which shows the desired estimate (6.2). Thus, the proof of Lemma 6.11 is complete. □

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \; \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 6.12. *There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,*

$$C^{-1}\, r^N \prod_{v \in R_+} \Big(L_{v,0} + r\Big)^{2k(v)} \;\leq\; \mu\big(B_\Omega(x_0, r)\big) \;\leq\; C\, r^N \prod_{v \in R_+} \Big(L_{v,0} + r\Big)^{2k(v)}, \tag{6.3}$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \qquad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\mathrm{dbl},\Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu\big(B_\Omega(x, 2r)\big) \;\leq\; C_{\mathrm{dbl},\Omega}\, \mu\big(B_\Omega(x, r)\big).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \qquad a_v := (a_{v,1}, \ldots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \qquad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \;\leq\; \|x - x_0\| \;\leq\; r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = \big|a_v \cdot (u - u_0)\big| \leq \|a_v\| \cdot \|u - u_0\| \leq L_*\, r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1\, (L_{v,0} + r), \qquad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2\sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu\big(B_\Omega(x_0, r)\big) = \int_{B_\Omega(x_0, r)} \phi(u)\, dx \leq C_2\, |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3\, r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2\, \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \qquad \eta := \frac{1}{4\big(\sqrt{N_0}\, \lambda + 1\big)}, \qquad \theta := \lambda\, \eta.$$

These parameters depend only on L_* and m_*, and satisfy

$$\theta > \eta > 0, \qquad \sqrt{N_0}\, \theta + \eta = \eta(\sqrt{N_0}\lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1,\dots,1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := \big(u_0 + \theta r\, \mathbf{1}_{N_0},\ z_0\big), \qquad B_c := B(x_c, \eta r).$$

We *claim* that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \le \|x - x_c\| + \|x_c - x_0\| \le \eta r + \theta r\sqrt{N_0} \le r,$$

since $\sqrt{N_0}\theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \le \|x - x_c\| \le \eta r$. This implies that for each $1 \le i \le N_0$,

$$u_i \ge (u_c)_i - \eta r = (u_0)_i + (\theta - \eta) r \ge 0,$$

since $u_0 \in \mathbb{R}^{N_0}_+$ and $\theta \ge \eta$. Thus $u \in \mathbb{R}^{N_0}_+$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c. Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r\, \mathbf{1}_{N_0} + \Delta_u, \qquad z = z_0 + \Delta_z, \qquad \|\Delta_u\|^2 + \|\Delta_z\|^2 \le (\eta r)^2,$$

so in particular $\|\Delta_u\| \le \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r\, a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\ge L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\|\,\|\Delta_u\| \ge L_{v,0} + r K_0 \ge c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \qquad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \ge c_0^{2\sum_{v\in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu\big(B_\Omega(x_0, r)\big) \ge \mu(B_c) = \int_{B_c} \phi(u)\, dx \ge C_4\, |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5\, r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4\, \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu\big(B_\Omega(x_0, 2r)\big) \le C_3\, (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \le C_3\, 2^{N+\mathcal{K}}\, r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \le C_3 C_5^{-1}\, 2^{N+\mathcal{K}}\, \mu\big(B_\Omega(x_0, r)\big),$$

where we set $\mathcal{K} := 2\sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\mathrm{dbl},\Omega} := C_3 C_5^{-1}\, 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 6.12. □

Let $\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$c_\Phi\, \|x - y\| \ \le\ \|\Phi(x) - \Phi(y)\| \ \le\ C_\Phi\, \|x - y\|. \tag{6.4}$$

Denote by $B_\Gamma(x,r) := \{y \in \Gamma : \ \|y-x\| < r\}$. It follows from (6.4) that for any $x_0 \in \Gamma$ and $r > 0$,

$$B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0,r)) \subset B_\Omega(\Phi(x_0), C_\Phi r). \tag{6.5}$$

This, together with Lemma 6.12, yields

$$\mu(\Phi(B_\Gamma(x_0,r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)). \tag{6.6}$$

Lemma 6.13. *For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis–parallel rectangle R' on Ω such that*

$$B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r). \tag{6.7}$$

Furthermore, we have

$$\mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)). \tag{6.8}$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis–parallel rectangle

$$R' := \Big(\prod_{i=1}^{N_0} \big[\max\{0, (u_0)_i - C_\Phi r\},\ (u_0)_i + C_\Phi r\big]\Big) \times \Big(\prod_{j=1}^{N-N_0} \big[(z_0)_j - C_\Phi r,\ (z_0)_j + C_\Phi r\big]\Big). \tag{6.9}$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N}\, C_\Phi r).$$

The second statement follows from the first one together with Lemma 6.12. The proof of Lemma 6.13 is finished. □

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 6.14. *There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,*

$$C^{-1}\, r^N \prod_{v \in R_+} (|\langle x, v\rangle| + r)^{2k(v)} \ \leq\ w\big(B_\Gamma(x,r)\big) \ \leq\ C\, r^N \prod_{v \in R_+} (|\langle x, v\rangle| + r)^{2k(v)}. \tag{6.10}$$

In particular, there exists a constant $C_{\mathrm{dbl},\Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w\big(B_\Gamma(x, 2r)\big) \ \leq\ C_{\mathrm{dbl},\Gamma}\, w\big(B_\Gamma(x,r)\big).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$w(B_\Gamma(x,r)) \sim w(B(x,r)). \tag{6.11}$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$w\big(B_\Gamma(x,r)\big) = \mu\big(\Phi(B_\Gamma(x,r))\big). \tag{6.12}$$

Applying μ to the inclusions (6.5) and using (6.12) yields

$$\mu\big(B_\Omega(\Phi(x), c_\Phi r)\big) \ \leq\ w\big(B_\Gamma(x,r)\big) \ \leq\ \mu\big(B_\Omega(\Phi(x), C_\Phi r)\big). \tag{6.13}$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (6.13) with Lemma 6.12 and the equality (3.6), we obtain that

$$w\big(B_\Gamma(x,r)\big) \;\simeq\; r^N \prod_{v\in R_+} \big(L_{v,0} + r\big)^{2k(v)} = r^N \prod_{v\in R_+} \big(|\langle x, v\rangle| + r\big)^{2k(v)}. \tag{6.14}$$

This proves (6.10). The second statement follows immediately from (6.10). The third statement follows from the inequalities (6.10) and (2.1). This completes the proof of Theorem 6.14. □

The following theorem demonstrates that the metric measure space $(\Gamma, \|\cdot\|, dw)$ supports $(1,1)$–Poincaré inequality.

[Zhijie will try to extends the $(1,1)$–Poincaré inequality to something like (p,q)–Poincaré inequality with some minor modifications, due to its independent interest in harmonic analysis]

Theorem 6.15. *There exist constants $\lambda \geq 1$ and $C > 0$ such that for every $x_0 \in \Gamma$, $r > 0$, and every Lipschitz function $f : \Gamma \to \mathbb{C}$,*

$$\fint_{B_\Gamma(x_0,r)} \big|f(x) - \fint_{B_\Gamma(x_0,r)} f(y)dw(y)\big|\, dw(x) \;\leq\; C\, r \fint_{B_\Gamma(x_0,\lambda r)} |\nabla f(z)|\, dw(z). \tag{6.15}$$

Proof. By decomposing $f = \operatorname{Re} f + i \operatorname{Im} f$ and then using triangular inequality, it suffices to show that for every $x_0 \in \Gamma$, $r > 0$, and every Lipschitz function $f : \Gamma \to \mathbb{R}$,

$$\fint_{B_\Gamma(x_0,r)} \fint_{B_\Gamma(x_0,r)} \big|f(x) - f(y)\big|\, dw(x)\, dw(y) \;\leq\; C\, r \fint_{B_\Gamma(x_0,\lambda r)} |\nabla f(z)|\, dw(z). \tag{6.16}$$

To show this inequality, we first let $g := f \circ \Phi^{-1}$ on Ω. It follows that $g \in \operatorname{Lip}(\Omega)$ with

$$|\nabla g(x)| \leq \|D\Phi^{-1}\|_{\mathrm{op}}\, |\nabla f(\Phi^{-1}x)| \qquad \text{for } x \in \Omega. \tag{6.17}$$

For all $x_0 \in \Gamma$ and $r > 0$, we let R' be the axis–parallel rectangle given in Lemma 6.13. Then

$$\begin{aligned}
&\fint_{B_\Gamma(x_0,r)} \fint_{B_\Gamma(x_0,r)} |f(x) - f(y)|\, dw(x)\, dw(y) \\
\text{(By change of variables)} &= \fint_{\Phi(B_\Gamma(x_0,r))} \fint_{\Phi(B_\Gamma(x_0,r))} |g(x) - g(y)|\, d\mu(x)\, d\mu(y) \\
\text{(By (6.5) and (6.6))} &\lesssim \fint_{B_\Omega(\Phi(x_0),C_\Phi r)} \fint_{B_\Omega(\Phi(x_0),C_\Phi r)} |g(x) - g(y)|\, d\mu(x)\, d\mu(y) \\
\text{(By (6.7) and (6.8))} &\lesssim \fint_{R'} \fint_{R'} |g(x) - g(y)|\, d\mu(x)\, d\mu(y) \\
\text{(By Lemma 6.11)} &\lesssim \operatorname{diam}(R') \fint_{R'} |\nabla g(x)|\, d\mu(x) \\
\text{(By (6.17))} &\lesssim \operatorname{diam}(R') \fint_{\Phi^{-1}(R')} |\nabla f(x)|\, dw(x) \\
\text{(By (6.7))} &\lesssim r \fint_{\Phi^{-1}(R')} |\nabla f(x)|\, dw(x).
\end{aligned} \tag{6.18}$$

To continue, we apply the inclusion (6.7) to see that for every $\xi \in R'$ we have $\|\xi - \Phi(x_0)\| \le \sqrt{N} C_\Phi r$. Combining with the inequality (6.4), we deduce that

$$\|\Phi^{-1}(\xi) - x_0\| \le \frac{1}{c_\Phi} \|\xi - \Phi(x_0)\| \le \frac{\sqrt{N} C_\Phi}{c_\Phi} r.$$

Since $R' \subset \Omega$ implies $\Phi^{-1}(R') \subset \Gamma$, we see that

$$\Phi^{-1}(R') \subset B_\Gamma(x_0, \lambda r), \qquad \text{where } \lambda := \frac{\sqrt{N} C_\Phi}{c_\Phi} \ge 1. \tag{6.19}$$

Moreover, it follows from (6.5) and (6.7) that $B_\Gamma(x_0, r) \subset \Phi^{-1}(R')$. Therefore,

$$\begin{aligned}
\fint_{\Phi^{-1}(R')} |\nabla f(x)|\, dw(x) &= \frac{1}{w(\Phi^{-1}(R'))} \int_{\Phi^{-1}(R')} |\nabla f(x)|\, dw(x) \\
\text{(By (6.7) and (6.19))} &\le \frac{1}{w(B_\Gamma(x_0, r))} \int_{B_\Gamma(x_0, \lambda r)} |\nabla f(x)|\, dw(x) \\
\text{(By Theorem 6.14)} &\lesssim \frac{1}{w(B_\Gamma(x_0, \lambda r))} \int_{B_\Gamma(x_0, \lambda r)} |\nabla f(x)|\, dw(x).
\end{aligned}$$

Plugging this inequality into (6.18) yields

$$\fint_{B_\Gamma(x_0,r)} \fint_{B_\Gamma(x_0,r)} |f(x) - f(y)|\, dw(x)\, dw(y) \lesssim r \fint_{B_\Gamma(x_0,\lambda r)} |\nabla f(x)|\, dw(x),$$

This completes the proof of Theorem 6.15. □

7. Failure of global Poincare inequality

Proposition 7.1. *Let $k > 0$ and let $w_k(x) = |x|^{2k}$. For every $N \in \mathbb{N}$, there exists smooth real-valued function f such that*

$$\inf_{c \in \mathbb{R}} \int_{-1}^{1} |f_N(x) - c| \cdot |x|^{2k} dx \ge \frac{1}{6(2k+1)},$$

$$\int_{\mathbb{R}} |f_N'(z)| \cdot |z|^{2k} dz = N^{-2k} \int_{\mathbb{R}} \frac{|z|^{2k} dz}{\cosh^2(z)}.$$

Proof. Set $f_N(x) = \tanh(Nx)$, $x \in \mathbb{N}$. Suppose $c \ge 0$. We have

$$\begin{aligned}
\int_{-1}^{1} |f_N(x) - c| \cdot |x|^{2k} dx &\ge \int_{-1}^{0} (c - \tanh(Nx)) |x|^{2k} dx = \\
= \frac{c}{2k+1} + \int_0^1 \tanh(Nx) x^{2k} dx &\ge \int_{\frac{1}{2}}^{1} \tanh(Nx) x^{2k} dx \ge \\
&\ge \tanh(\frac{1}{2}N) \cdot \int_{\frac{1}{2}}^{1} x^{2k} dx \ge \frac{1}{3} \cdot \frac{1}{4k+2}.
\end{aligned}$$

Suppose $c < 0$. We have

$$\begin{aligned}
\int_{-1}^{1} |f_N(x) - c| \cdot |x|^{2k} dx &\ge \int_0^1 (\tanh(Nx) - c) |x|^{2k} dx = \\
= -\frac{c}{2k+1} + \int_0^1 \tanh(Nx) x^{2k} dx &\ge \int_{\frac{1}{2}}^{1} \tanh(Nx) x^{2k} dx \ge
\end{aligned}$$

$$\geq \tanh(\frac{1}{2}N) \cdot \int_{\frac{1}{2}}^{1} x^{2k} dx \geq \frac{1}{3} \cdot \frac{1}{4k+2}.$$

At the same time,

$$\int_{\mathbb{R}} |f_N'(z)| \cdot |z|^{2k} dz = \int_{\mathbb{R}} \frac{N|z|^{2k} dz}{\cosh^2(Nz)} = N^{-2k} \int_{\mathbb{R}} \frac{|z|^{2k} dz}{\cosh^2(z)}.$$

□

8. Constant characterization

In this section, we provide a proof for Theorem 1.3. The sufficient condition is obvious, since $[b, R_{\mathrm{Dunkl},j}] = 0$ when b is a constant. Thus, it suffices to show the necessary condition. It suffices to consider the critical case $p = N$, by the inclusion $S^p \subset S^q$ for $p < q$.

Notation 8.1. *Let $\Gamma \subset \mathbb{R}^N$ be a Weyl chamber so that (by Lemma 3.3)*

$$\Gamma = \Big\{ x \in \mathbb{R}^N : \ \langle x, \beta \rangle \geq 0 \text{ for all } \beta \in \Delta \Big\},$$

where $\Delta = \{\alpha_1, \ldots, \alpha_{N_0}\}$ is the simple system associated with Γ. For any $\alpha \in \Delta$, let

$$F_\alpha := \Big\{ x \in \mathbb{R}^N : \ \langle x, \alpha \rangle = 0, \ \langle x, \beta \rangle \geq 0 \text{ for all } \beta \in \Delta \setminus \{\alpha\} \Big\}$$

be the facet of Γ associated with α. Moreover, let

$$\mathrm{relint}(F_\alpha) := \Big\{ x \in \mathbb{R}^N : \ \langle x, \alpha \rangle = 0, \ \langle x, \beta \rangle > 0 \text{ for all } \beta \in \Delta \setminus \{\alpha\} \Big\}$$

be the relative interior of F_α (taken in its affine hull).

Lemma 8.2. *Let $\Gamma \subset \mathbb{R}^N$ be a Weyl chamber. For any $\alpha \in \Delta$, we have*

$$\mathrm{relint}(F_\alpha) \neq \varnothing.$$

Proof. Write $\alpha = \alpha_i$ for some $i \in \{1, \ldots, N_0\}$. Let $G = (\langle \alpha_j, \alpha_k \rangle)_{j,k=1}^{N_0}$ be the Gram matrix, which is invertible since Δ is a basis of W (see Lemma 3.3). Let $\varrho \in \mathbb{R}^{N_0}$ be defined by $\varrho_i = 0$ and $\varrho_j = 1$ for $j \neq i$. Set $\upsilon = G^{-1}\varrho$. Define $z := \sum_{j=1}^{N_0} \upsilon_j \alpha_j \in W \subset \mathbb{R}^N$. Then for each $k = 1, \ldots, N_0$,

$$\langle z, \alpha_k \rangle = \sum_{j=1}^{N_0} \upsilon_j \langle \alpha_j, \alpha_k \rangle = (G\upsilon)_k = \varrho_k.$$

In particular, $\langle z, \alpha \rangle = \langle z, \alpha_i \rangle = \varrho_i = 0$ and $\langle z, \beta \rangle = 1 > 0$ for all $\beta \in \Delta \setminus \{\alpha\}$. Hence, $z \in \mathrm{relint}(F_\alpha)$, which implies that $\mathrm{relint}(F_\alpha) \neq \varnothing$. □

Lemma 8.3. *Let $\Gamma \subset \mathbb{R}^N$ be a Weyl chamber. For each $\alpha \in \Delta$, let z_α be an arbitrary point in* $\mathrm{relint}(F_\alpha)$. *There exists $\delta_\alpha > 0$ such that, with the notation*

$$U_\alpha := B(z_\alpha, \delta_\alpha), \qquad U_\alpha^+ := \{x \in U_\alpha : \langle x, \alpha \rangle > 0\}, \qquad U_\alpha^- := \{x \in U_\alpha : \langle x, \alpha \rangle < 0\},$$

one has

$$\Gamma \cap U_\alpha = U_\alpha^+ \cup (F_\alpha \cap U_\alpha) \quad \text{and} \quad \sigma_\alpha(\Gamma) \cap U_\alpha = U_\alpha^- \cup (F_\alpha \cap U_\alpha). \tag{8.1}$$

Proof. Fix $\alpha \in \Delta$. Note that for any $\varpi \in R_+\backslash\{\alpha\}$, it follows from Lemma 3.3 that $\varpi = \sum_{\beta\in\Delta} c_\beta \beta$ for some $c_\beta \geq 0$. Moreover, there exists some $\beta \neq \alpha$ with $c_\beta > 0$, and hence,

$$\langle z_\alpha, \varpi\rangle = \sum_{\beta\neq\alpha} c_\beta \langle z_\alpha, \beta\rangle > 0.$$

Similarly, for any $\varpi \in R_-\backslash\{-\alpha\}$, we have $\langle z_\alpha, \varpi\rangle < 0$. Therefore, $\langle z_\alpha, \varpi\rangle \neq 0$ for every $\varpi \in R\backslash\{\pm\alpha\}$. Take

$$\delta_\alpha := \min_{\varpi\in R\backslash\{\pm\alpha\}} \frac{|\langle z_\alpha, \varpi\rangle|}{2\|\varpi\|} \ > \ 0.$$

Note that for any $x \in \mathbb{R}^N$ with $\|x - z_\alpha\| < \delta_\alpha$ and any $\varpi \in R \setminus \{\pm\alpha\}$, by the Cauchy–Schwarz inequality we have

$$\big|\langle x, \varpi\rangle - \langle z_\alpha, \varpi\rangle\big| = \big|\langle x - z_\alpha, \varpi\rangle\big| \leq \|x - z_\alpha\|\,\|\varpi\| < \delta_\alpha\,\|\varpi\| \leq \frac{1}{2}\,|\langle z_\alpha, \varpi\rangle|,$$

and therefore, $\langle x, \varpi\rangle$ has the same sign as $\langle z_\alpha, \varpi\rangle$.

In particular, since $z_\alpha \in \operatorname{relint}(F_\alpha)$, we have $\langle z_\alpha, \beta\rangle > 0$ for all $\beta \in \Delta \setminus \{\alpha\}$, and thus $\langle x, \beta\rangle > 0$ for all $x \in U_\alpha$. Hence,

$$\Gamma \cap U_\alpha = \{x \in U_\alpha : \langle x, \alpha\rangle \geq 0\} = U_\alpha^+ \cup (F_\alpha \cap U_\alpha).$$

Applying σ_α to the equality above and using $\sigma_\alpha(U_\alpha) = U_\alpha$, $\sigma_\alpha(U_\alpha^+) = U_\alpha^-$ and $\sigma_\alpha(F_\alpha) = F_\alpha$, we obtain the second identity in (8.1). This completes the proof of Lemma 8.3. □

Lemma 8.4. *Let $\Gamma \subset \mathbb{R}^N$ be a Weyl chamber. Then for each $\alpha \in \Delta$, there exist two families of balls $\{B_m\}_{m\geq 1} \subset \Gamma$ and $\{\hat{B}_m\}_{m\geq 1} \subset \sigma_\alpha(\Gamma)$ such that each family is pairwise disjoint, and such that for each $m \geq 1$, the balls B_m and $\hat{B}_m$ have the same radius $r_m > 0$. Moreover, for every $m \geq 1$ and every $(x, y) \in B_m \times \hat{B}_m$,*

$$11r_m \leq \|x - y\| \leq 15r_m. \tag{8.2}$$

Proof. We will use the notation in Lemma 8.3. For each $m \geq 1$, we choose

$$B_m := B(x_m, r_m), \quad \hat{B}_m := B(y_m, r_m)$$

with

$$x_m := z_\alpha + \frac{9r_m}{\sqrt{2}}\alpha,\ y_m := z_\alpha - \frac{4r_m}{\sqrt{2}}\alpha,\ r_m := 2^{-m-4}\delta_\alpha.$$

Next, we show that $\{B_m\}_{m=1}^\infty$ and $\{\hat{B}_m\}_{m=1}^\infty$ satisfy all the desired properties.

• **Step 1.** *Verify that $\{B_m\}_{m=1}^\infty \subset \Gamma$, $\{\hat{B}_m\}_{m=1}^\infty \subset \sigma_\alpha(\Gamma)$.*

Since

$$\|x_m - z_\alpha\| + r_m = 10r_m < \delta_\alpha, \quad \|y_m - z_\alpha\| + r_m = 5r_m < \delta_\alpha,$$

we have $B_m \subset U_\alpha$ and $\hat{B}_m \subset U_\alpha$. Moreover, note that for any $x \in B_m$,

$$\langle x, \alpha\rangle \geq \langle x_m, \alpha\rangle - \sqrt{2}\|x - x_m\| \geq 9\sqrt{2}r_m - \sqrt{2}r_m = 8\sqrt{2}r_m > 0.$$

This, together with Lemma 8.3, yields $x \in U_\alpha^+ \subset \Gamma$. Hence, $\{B_m\}_{m=1}^\infty \subset \Gamma$. Similarly, for any $y \in \hat{B}_m$,

$$\langle y, \alpha\rangle \leq \langle y_m, \alpha\rangle + \sqrt{2}\|y - y_m\| \leq -4\sqrt{2}r_m + \sqrt{2}r_m = -3\sqrt{2}r_m < 0.$$

This, together with Lemma 8.3, yields $y \in U_\alpha^- \subset \sigma_\alpha(\Gamma)$. Hence, $\{\hat{B}_m\}_{m=1}^\infty \subset \sigma_\alpha(\Gamma)$.

• **Step 2.** *Verify that $\{B_m\}_{m=1}^\infty$ and $\{\hat{B}_m\}_{m=1}^\infty$ are two families of pairwise disjoint balls.*

If $m < n$, then

$$\|x_m - x_n\| = 9|r_m - r_n| = 9r_m(1 - 2^{m-n}) \geq \frac{9}{2}r_m,$$

$$\|y_m - y_n\| = 4|r_m - r_n| = 4r_m(1 - 2^{m-n}) \geq 2r_m,$$

$$r_m + r_n = r_m(1 + 2^{m-n}) \leq \frac{3}{2}r_m.$$

Combining these inequalities, we obtain that $\|x_m - x_n\| > r_m + r_n$ and $\|y_m - y_n\| > r_m + r_n$, which implies that $B_m \cap B_n = \varnothing$ and $\hat{B}_m \cap \hat{B}_n = \varnothing$.

• **Step 3.** *Verify the inequality* (8.2).

Indeed, this can be obtained by the triangle inequality as follows:

$$\|x - y\| \leq \|x_m - y_m\| + \|x - x_m\| + \|y - y_m\| \leq 13r_m + r_m + r_m = 15r_m,$$

$$\|x - y\| \geq \|x_m - y_m\| - \|x - x_m\| - \|y - y_m\| \geq 13r_m - r_m - r_m = 11r_m.$$

This finishes the proof of Lemma 8.4. □

[Zhijie will write the concept of lower dimension and $(1, d)$–Poincaré inequality later]

Lemma 8.5. [9, Corollary 4.3] *Let (X, ρ, μ) be a doubling metric measure space having lower dimension d and supporting the $(1, d)$–Poincaré inequality. If $f \in L^1_{\mathrm{loc}}(X, \mu)$ satisfies*

$$\left[(x, y) \mapsto \frac{|f(x) - f(y)|^d}{\mu(B(x, \rho(x, y)))^2}\right] \in L^1_{\mathrm{loc}}(X \times X, \mu \times \mu), \tag{8.3}$$

then f is equal to a constant almost everywhere on X.

Proposition 8.6. *Suppose $N \geq 2$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. If $[b, R_{\mathrm{Dunkl},j}] \in S^N$ for some $j \in \{1, 2, ..., N\}$, then b is a constant on $\mathbb{R}^N$ almost everywhere.*

Proof. The proof is divided into three steps.

• **Step 1.** *Show that $b \equiv c_\Gamma$ almost everywhere for some constant c_Γ on each Weyl chamber Γ.*

By Lemma 5.6, if $[b, R_{\mathrm{Dunkl},j}] \in S^N$ for some $j \in \{1, 2, ..., N\}$, then $b \in B^N(\mathbb{R}^N, dw)$. This, together with the inequality (6.11), yields that for every Weyl chamber Γ, we have

$$\|b\|_{B^N(\Gamma, dw)} := \left(\int_\Gamma \int_\Gamma \frac{|b(x) - b(y)|^N}{w(B_\Gamma(x, \|x - y\|))^2} dw(x) dw(y)\right)^{1/N} < +\infty.$$

By Theorem 6.14, $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space having lower dimension N. Together with Theorem 6.15 and Hölder's inequality, this implies that $(\Gamma, \|\cdot\|, dw)$ supports a $(1, N)$–Poincaré inequality. It follows from Lemma 8.5 that $b \equiv c_\Gamma$ on Γ almost everywhere for some constant c_Γ.

• **Step 2.** *Show that $c_\Gamma = c_{\sigma_\alpha(\Gamma)}$ for each Weyl chamber Γ and $\alpha \in \Delta$.*

Indeed, if this is not the case, then $A := c_\Gamma - c_{\sigma_\alpha(\Gamma)} \neq 0$ for some Weyl chamber Γ and some $\alpha \in \Delta$(the simple system associated with Γ). Let $\{(B_m, \hat{B}_m)\}_{m \geq 1}$ be the family of pairs of balls constructed in Lemma 8.4. By (8.2) and the doubling inequality (2.2), for all $(x, y) \in B_m \times \hat{B}_m$,

$$w(B(x, \|x - y\|)) \sim w(B_m) \sim w(\hat{B}_m).$$

It follows that

$$I_m := \int_{B_m}\int_{\hat{B}_m} \frac{|b(x)-b(y)|^N}{w(B(x,\|x-y\|))^2} dw(x)dw(y) \gtrsim A^N.$$

Since for any $m \neq n$, $B_m \cap B_n = \varnothing$ and $\hat{B}_m \cap \hat{B}_n = \varnothing$, we have

$$\|b\|^N_{B^N(\mathbb{R}^N,dw)} \geq \sum_{m=1}^{\infty} I_m \gtrsim \sum_{m=1}^{\infty} A^N = +\infty.$$

This is a contradiction to $b \in B^N(\mathbb{R}^N, dw)$. Therefore, $c_\Gamma = c_{\sigma_\alpha(\Gamma)}$.

• **Step 3.** *Show that $c_{\Gamma_1} = c_{\Gamma_2}$ for any two distinct Weyl chambers Γ_1 and Γ_2.*

Fix two Weyl chambers Γ_1 and Γ_2. By Lemma 3.10, there exist an integer $M \geq 0$ and roots $\alpha_1, \dots, \alpha_M \in R$ such that, if we define $\Gamma^{(0)} := \Gamma_2$ and

$$\Gamma^{(k)} := \sigma_{\alpha_k}\big(\Gamma^{(k-1)}\big), \qquad k = 1, \dots, M,$$

then $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$, and $\Gamma^{(M)} = \Gamma_1$.

Applying Step 2 to the Weyl chamber $\Gamma^{(k-1)}$ and the simple root $\alpha_k \in \Delta(\Gamma^{(k-1)})$, we obtain

$$c_{\Gamma^{(k)}} = c_{\sigma_{\alpha_k}(\Gamma^{(k-1)})} = c_{\Gamma^{(k-1)}}, \qquad k = 1, \dots, M.$$

By induction, $c_{\Gamma^{(M)}} = c_{\Gamma^{(0)}}$, i.e., $c_{\Gamma_1} = c_{\Gamma_2}$.

This completes the proof of Proposition 8.6. □

9. Quantum differentiability in the Dunkl setting

(maybe)Zhijie will do this after the confirmation of the main body.

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References

[1] S.G. Bobkov and C. Houdré. Some connections between isoperimetric and Sobolev-type inequalities. *Mem. Amer. Math. Soc.*, 129(616):viii+111, 1997. 36

[2] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. 7

[3] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. 7

[4] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at https://arxiv.org/abs/2403.08249, 2024. 7

[5] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. 4

[6] G. Heckman. Coxeter groups. Lecture notes available at https://www.math.ru.nl/~heckman/CoxeterGroups.pdf. Radboud University Nijmegen, 2018. 8, 9, 11

[7] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. 4

[8] T. Hytönen and R. Korte. Schatten properties of commutators and equivalent Sobolev norms on metric spaces. Preprint available at https://arxiv.org/abs/2411.02613, 2024. 7, 8, 12, 13, 25

[9] T. Hytönen and R. Korte. Characterisations of Sobolev spaces and constant functions over metric spaces. Preprint available at https://arxiv.org/abs/2411.02613, 2025. 44

[10] S. Janson and T.H. Wolff. Schatten classes and commutators of singular integral operators. *Ark. Mat.*, 20(2):301–310, 1982. 12

[11] J. Journé. *Calderón-Zygmund operators, pseudodifferential operators and the Cauchy integral of Calderón*, volume 994 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, 1983. 35

[12] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. 7

[13] J. Li, J. Pipher, and L.A. Ward. Dyadic structure theorems for multiparameter function spaces. *Rev. Mat. Iberoam.*, 31(3):767–797, 2015. 6

[14] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. 7

[15] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. 7, 8

[16] S. Vempala. Geometric random walks: a survey. In *Combinatorial and computational geometry*, volume 52 of *Math. Sci. Res. Inst. Publ.*, pages 577–616. Cambridge Univ. Press, Cambridge, 2005. 34

[17] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at https://arxiv.org/abs/2411.05810, 2024. 7, 16

[18] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at https://arxiv.org/abs/2407.14988, 2024. 7

Zhijie Fan, School of Mathematical Sciences, South China Normal University, Guangzhou 510631, China
E-mail address: fanzhj3@mail2.sysu.edu.cn

Ji Li, Department of Mathematics, Macquarie University,NSW2109, Australia
E-mail address: ji.li@mq.edu.au

Dmitriy Zanin, University of New South Wales, Kensington, NSW 2052, Australia
E-mail address: d.zanin@unsw.edu.au